PART VI

ON THE FOUNDATION OF PSYCHOPHYSIOLOGY

It is an important principle of physiological epistemology that a phenomenon which occurs generally, cannot possibly be the *specific* function of an organ which is peculiar to a few forms only. (306) JACQUES LOEB

I have observed, as a consequence of serious and repeated emotional shocks, very curious cases of infantilism in adults, where complete amnesia, accompanying sexual inhibition and disturbances of the affective area, produced the mentality and conduct of a little child. (411) HENRI PIÉRON

The organism is reacting as a whole to its environment as a whole, and it is doing so in ways that cannot be formulated in terms of an algebraic sum or simple mechanical resultant of the interplay of the simple reflex responses to external stimulation. . . . the mechanisms of traditional reflexology seem hopelessly inadequate. (224) C. JUDSON HERRICK

CHAPTER XX

GENERAL CONSIDERATIONS

We know also how different extra stimuli inhibit and discoordinate a well-established routine of activity, and how a change in a pre-established order dislocates and renders difficult our movements, activities and the whole routine of life. (394) I. P. PAVLOV

The experiments show that a compound stimulus the component units of which remain themselves unaltered, and consequently most probably affect the same cells of the cerebral cortex, behaves in different modifications as a different stimulus, evoking in these cells now an excitatory process and now an inhibitory one. (394) I. P. PAVLOV

We thus come to the following conclusion: when perfectly neutral stimuli fall upon the hemispheres at a time when there prevails a state of inhibition they acquire an inhibitory function of their own, so that when they act subsequently upon any region of the brain which is in a state of excitation they produce inhibition. (394) I. P. PAVLOV

Some of the most important researches in the function of the higher nervous centres have been done lately by Professor Pavlov in his work on the so-called 'conditioned reflexes'. This work was developed in a series of papers covering a period of nearly thirty years of experimentation, but the average international scientist did not know this work as an entirety, because the papers were scattered and written mostly in Russian. Only in 1927 did the Oxford Press publish Pavlov's *Conditioned Reflexes, an Investigation of the Physiological Activity of the Cerebral Cortex* in the English translation of Doctor G. V. Anrep; and in 1928 The International Publishers (New York) published Pavlov's *Lectures on Conditioned Reflexes, Twenty-five Years of Objective Study of the Higher Nervous Activity (Behaviour) of Animals* in the translation of Doctor W. Horsley Gantt. Both translators were collaborators of Professor Pavlov in Leningrad for a number of years. In these two books, the latest experiments and interpretations are given.

Hitherto, most of the researches on the function of the higher nervous systems were formulated in 'psychological' languages, which, obviously, are not fit for physiological disciplines. Professor Pavlov, himself, suggests this fact as an explanation why, until his work, the physiology of the cerebral cortex was so little known. There is no doubt that the descriptive physiological language of happenings, functionings., used by him exclusively, is responsible for his results. This language suggests structurally new experimentations, which suggestions are lacking in other accounts of the kind where antiquated 'psychological' terms are used.

Although I knew as much as the average scientist about the work of Pavlov, this knowledge was not integrated enough to make some

issues clear. But, after I had formulated my $\bar{A}$-system, I read the books of Pavlov and found, to my great satisfaction, that a neurological mechanism, the analysis of which underlies my own work, and the existence of which was independently discovered by me on *theoretical* grounds, had been discovered by Professor Pavlov and his co-workers on *experimental* grounds, thus supplying additional experimental verification for my system.

It seems that the so-called 'ethics'., in general, sanity, which underlie desirable human characteristics have a definite *physiological* mechanism, automatically involving on psycho-logical levels these desirable semantic attitudes. It appears that some of the psycho-logical problems enormously complex and difficult to reach, or even inaccessible, are solved, not by preaching, but by the most simple and elementary *physiological* training, a fact which has been verified empirically. Obviously, such simplification, if at all possible, must be of fundamental importance.

Physiology deals, in the main, with the functioning of organs in organisms, and results in various formulations. Thus, there might be an hypothetical 'physiological theory of most effective feeding', for instance, stating that food should be secured first in one's hand, or spoon, or fork, before putting it in the mouth,. A group of people who habitually disregarded the 'physiological theory' and abandoned attempts to act in accordance with it after the first unsuccessful one, would be badly underfed or would simply perish. Facts of experience show that some such 'physiological theory' must have been known and applied from time immemorial, and that, perhaps, because of it we survive at all!

How about the 'mental' field? As I demonstrate—and close observation will verify this very generally—the existing theories of 'mental' life, closely related with our linguistic habits, are *A*, grossly inadequate, and lead to a wholesale production of morons, imbeciles, 'emotionally' disturbed, and, in general, un-sane individuals. Investigation shows the possibility of a simple and obvious *physiological* theory of the use of our nervous system, which automatically leads to desirable psycho-logical, semantic states of general sanity.

In the frivolous example of a 'physiological theory of feeding' given above, the problems of *order* were important. In the physiological theory of sanity, order becomes paramount. Processes and function involve series of states, by necessity exhibiting order. Adjustment to life-conditions means adjustment of processes, and a physiological theory of sanity must be based structurally on four-dimensional order, where 'space' and 'time' are indivisibly interwoven.

Pavlov shows, in an unusually impressive variety and numbers of experiments, how 'order' and 'delay' (four-dimensional order, in the language used here in this connection) are intimately related with most fundamental processes in the higher nervous centres, and how, by the changes or interplays of them, we can produce or eliminate *pathological states* of the nervous system.

In the human field we find a quite similar situation, unanalysable by older methods, because all order involves asymmetrical relations, which, as we have already shown, cannot be dealt with by *A* means.

The issue is clear and definite: either we persist in our old *A* habits of speech, in which case asymmetrical relations and order evade our grasp, and proper evaluation and sanity are *physiologically* impossible, *or* we build a $\bar{A}$-system free, or at least more free, from these evaluational limitations, which allows us to deal with order, and sanity becomes *physiologically possible*.

'Stimuli are never "simple", and, by necessity, involve fourfold space-time structure and order. Survival values involve, also, this four-dimensional order. For instance, the natural survival order is "senses" first, "mind" next; object first, label next; description first, inference next,. The reversal of the natural order appears pathological and pathogenic and is found as a symptom in practically all forms of "mental" ills, as well as in most human difficulties and disturbances which, at present, are still not considered abnormal. Thus, objectivity is ascribed to words, "mind" projected into "senses", inferences evaluated as descriptions .,—quite common "symptoms" . . . Observations on human levels show that we still copy animals in our nervous responses, confuse orders of abstractions (non-existent for the animals), leading fatalistically to the reversal of the natural order and to pathological results, making the great majority of us un-sane.'*

A structural *non-el* enquiry into the objective world shows quite clearly that no event is ever 'simple'; it is, at least, a limited whole of interrelated factors. The eventual 'simplicity' is manufactured by a nervous process of higher and higher abstractions.

In our consideration of 'order' and 'delay', and the role they play in connection with the activities of the nervous system, we must first discriminate sharply between the objective level which is *un-speakable*, because anything that can be said *is not* the object, and the verbal level, on which we can, at will, concentrate attention on similarities, or differences, or both. Secondly, we must pay special attention to structure—

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*From—Discussion by A. Korzybski. *Proceedings of the First International Congress of Mental Hygiene*. New York, 1932.

that is to say, search for structure in the empirical world, and, once this has been found, adjust, accordingly, the structure of our language.

The structure of the daily, as well as of the 'philosophical', language, which we inherited, in the main, from our primitive ancestors, is such that we have *separate terms* for factors which are not separable, such as 'matter', 'space', 'time', or 'body', 'soul', 'mind',. Then, as it were, we try to make out of the word, flesh, by reversing the natural order and affectively ascribing a delusional *objectivity* to these terms.

If we deal with the silent, un-speakable, objective level and try to divide according to the implications of the verbal division, we find a brutal fact, which, until Einstein and Minkowski, has escaped scientific verbal formulation, that this cannot be done at all. On the objective level every dealing with 'matter' involves 'space' and 'time'; any dealing with 'space' involves some fulness of something and 'time'; and every dealing with 'time' involves 'something' and 'space'.

The structure of the world happens to be such that empirically 'matter', 'space', 'time', cannot be divided; wherefore, we should have a *non-el* language of *similar structure.* This was accomplished by Einstein-Minkowksi, when they created a language of 'space-time', in which the hard lumps against which we bump our noses are connected analytically with the curvature of space-time.

In this new, *non-el,* four-dimensional language, every three-dimensional point of 'space' has a date, and so is different. For our purpose, we do not need, at present, to bother much about its curvature or the kinks in space-time, called in the old way 'matter', but we must emphasize that the fourfold order is of great importance, as it corresponds structurally to *experience,* and is intimately connected with physiological reactions, the semantic included.

There is a great deal of confusion about these problems among laymen and also among scientists. From a structural point of view, the issues are quite simple, and there is nothing sensational in the latest announcement of Einstein that 'space' in its importance is displacing 'matter' (*Nottingham Lectures*). Naturally, the statement *in this form* is rather baffling and attracted much—even newspaper—attention. Yet it seems that even the einsteinists do not fully realize the *verbal, structural,* and semantic issues involved.

For the layman, as well as for the majority of the physicists in their less sober, or metaphysical, moments, 'space' is 'emotionally' newtonian and an 'absolute void', which, of course, being 'absolute nothingness', cannot have *objective* existence, by definition. For Einstein, 'space-time' is, semantically, 'fulness', not 'emptiness', and, in *his language,* he does

not need any term like 'ether', as his 'plenum', structurally, covers the ground, without his committing himself to a definite two-valued mechanistic ether. The confusion of orders of abstractions, from which we all suffer, is semantic, and is due to disregard of the structure and role of language. If we accept a *non-el* language of space-time, structurally we deal with fulness, and we should not use the term 'space', as its old semantic implications are 'emptiness' and so are very confusing. The 'sensation' of Einstein's declaration amounts to the fact that the submicroscopic fulness ('space') is more important than a few kinks or concentrations of that fulness ('matter'),—a fact which science has established, and which is quite obvious.

Experiments with 'conditioned reflexes' have established firmly the fact that stimuli can be compounded, and that, when established, the compound stimulus acts as a unit, and that a change in the four-dimensional order of factors (including delays) acts as a *different* stimulus, not necessarily resulting in the established reflex. This often introduces great complexities.

As an example of this, we will use the so-called 'delayed reflex'. When established, the 'conditioned reflex' does not appear at once after the stimulation, but after the stimulation combined with the usual 'time'-delay has occurred, thus showing that the 'time' factor plays a physiological role in a compound stimulus. Organisms live in, and consist of, periodic processes, such as the alternation of day and night, sleep, taking in food, heart-beats, breathing, electronic pulsations . ; so that *any* stimulus, no matter how nominally 'simple', is, in reality, a compound stimulus of, let us say, x and y heart-beats and what not. An organism represents, invariably, a clock of some sort, and, when that clock stops, life ceases.

Under such actual structural conditions a four-dimensional analysis makes every 'simple' stimulus compound, and thus four-dimensional *order* becomes a potent *physiological* factor, exerting definite effects. The interplay of four-dimensional order of factors represents, in general, a new stimulus ; we have an interplay of positive and negative excitations which may lead to clashes between the two that the nervous system finds difficult to resolve, and so pathological results follow.

If we pass to sub-microscopic levels and processes, we find that. although we may speak of them as 'chemical' or 'stimuli of greater physiological strength'., yet, by structural necessity, they represent different kinds of *multi-dimensional* order, because, as we say in 1933, the dynamic physical unit of that order is a quantum of action. The metaphysician should not get excited about this statement, because whatever

he might say will also be a verbal statement of a given date, made mostly without any structural considerations, and based mostly upon the reversed survival order, confusion of orders of abstractions and other semantic disturbances. As the world, both outside and inside our skins, is invariably found to exhibit a fourfold space-time order, it is inevitable that this order should be structurally impressed on the nervous system, establishing a natural survival order. Therefore, changes in this order on the macroscopic level, the level of outward events, must have direct inward sub-microscopic effects, disturbing or restoring the nervous equilibrium. This statement may appear innocent; it is not; it has a vital human significance, as it involves standards of evaluation. In short, it means that, in the *actual application* of the consideration of order in education and training on the daily-life levels, we can affect the evasive (as yet) microscopic and sub-microscopic structural levels of the human nervous system, thus directly affecting our *s.r* and behaviour.

To make this clearer, let us recall some of the neurological researches of Bolton (as quoted by Herrick). The cortex has different layers, characterized by the difference in the number, size, shape, internal structure, and density of neural cells. Bolton's third layer of granules divides the cortex into two types of layers. Those closer to the base of the brain, or below the third, are called the infragranular; those above, the supragranular, layers.

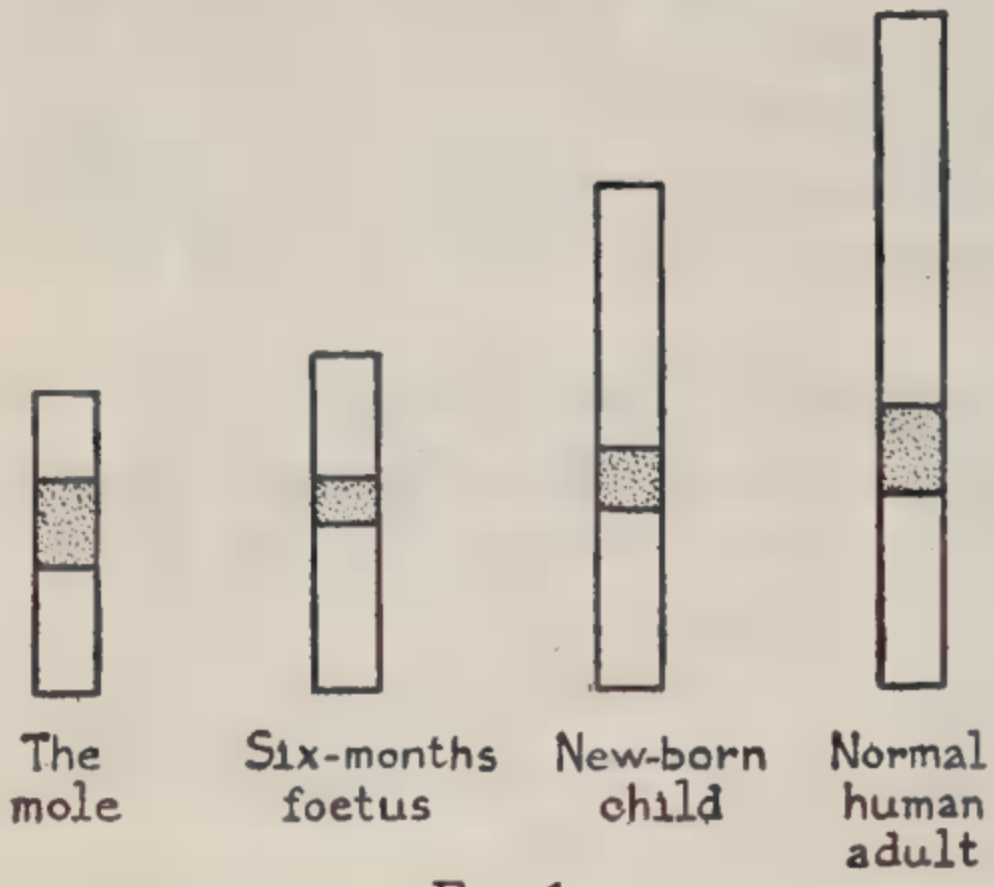

FIG. 1

Diagrams of the relative thickness of the supragranular, granular, and infragranular cerebral cortex in the six-months foetus, the new-born child, the normal human adult, and the adult mole. The granular layer is dotted. (Redrawn from G. A. Watson (1907), and adopted from Herrick.[1])

The lower mammals show a well-developed infragranular cortex, and a very poorly organized supragranular cortex, the latter increasing in relative size and complexity as we ascend the animal series. On the human level, we find a most important, and usually disregarded, fact—that the human nervous system is not completed at birth, but develops structurally years after birth.

The above explains why animalistic theories and methods, primitive-made languages of wrong structure, and similar relics, result in training the *s.r* of our children in the patho-

logical order, and bring about such great harm, individually and collectively. It enables us to understand, also, why all forms of 'mental' ills invariably exhibit infantile characteristics of some sort.

If we train a child with a physically undeveloped nervous system in animalistic doctrines strictly connected with a primitive-made language of wrong structure, in the pathological reversed order of responses, such semantic training must affect harmfully the still developing nervous system. So, when we say, and demonstrate, that we still copy animals in our nervous responses, we imply an undeveloped or thwarted nervous system, of which the development has been arrested or made regressive. Such a deficiency, of course, is superimposed functionally, and so structurally, upon whatever congenital deficiency there might have been in a given case. We are nearly all in a situation of this kind. We continue to be educated under animalistic conditions since we became time-binders, which, from a biological point of view, is a very recent event, and it is not rash to assume that our nervous system is still not fully developed, the more so that we submit the cortex, which in childhood is still incomplete, to injurious semantic influences. Obviously, such a fundamental human function as language, when used in a way not in correspondence with the structure of the nervous system, must act detrimentally on its development.

In congenital extreme imbecility, the cortex is poorly organized, thin, and deficient in nerve cells, and the infragranular layers show less impairment than the supragranular layers. It seems that Bolton's second supragranular layer is the last to mature, and its relative development corresponds to the relative development of an animal or human being, and, in a way, it goes parallel with the so-called 'intelligence'.

In human defectives, its deficiency corresponds with the degree of psycho-logical arrest, regression, or deterioration. Let us recall that these 'mental' deficiencies, which, in behaviour as well as in nervous structure, take us one step (or several) back toward the level of the infant, or even to that of the animal, are always connected with infantile behaviour in adults, and semantic disturbances.

Nervous as well as muscular tissues have differentiated from the general protoplasm, and we know positively that, through training, we can enlarge or otherwise improve muscular tissues, and there is no reason to doubt that something similar can be done to nervous tissue. All education, and the establishing of any conditional or *s.r*, shows this, although in a rather vague way.

If, by a *physiological* training based on order, we can alter a nervous deficiency, as shown by behaviour, we may conclude that there are physio-

logical means by which we can effectively train and help the development of nervous cells and supragranular layers—or, at least, *not hinder* their natural development. On the macroscopic levels, this beneficial training consists in forming habits of proper evaluation through the natural survival order. The effect of this on the sub-microscopic level is neural, colloidal, and structural, a result which, by the older methods, could not be reached, either with ease or with effectiveness.

Although these conclusions are necessary, it is impossible directly to verify them empirically, because we should have to dissect the brain of a given person before, and during, the training. In this case, as in many others, we have to observe 'human nature' and semantic responses to stimuli of a given individual before and after the training; and, on the foundation of what we know of the development of the nervous system in animals, infants, 'mentally' defective and well-developed adults, build our eventual conclusion as preliminary hypotheses for further structural testing, improvement, and empirical verification.

To realize fully the importance and necessity of this conclusion, we must first understand that, in accordance with the modern discoveries of mathematics, physics, chemistry, colloidal chemistry, and other branches of science, all 'function' depends on *structure,* because the unit-brick of structure represents a dynamic unit of a quantum of action. In the remarks which follow, it is impossible to be as full and precise as I should like to be, because to be able to do so would be equivalent to being able to solve all scientific problems; yet the reader should realize that the considerations of structure will become extremely creative and helpful as long as we recognize a quantum of action, or any other *dynamic unit.*

We may recall that the characteristics of molecules are due to atomic structure, and that the characteristics of atoms are due to electronic structure. The latest quantum theories also seem to find that the characteristics of electrons are an outgrowth of structure; and, if the suggestions of Dirac are verified, even the difference between positive and negative electricity is structural. Even at present it appears that 'structure' is not only a term fit to handle and explain, but that it has an objective counterpart, allowing a similarity of structure, and, therefore, making the understanding of this world possible.

This last very important semantic point is based on the fact that relations of similar structure have similar 'logical' characteristics, and that, therefore, in structure we find means by which the events can always be made intelligible to us, and so properly evaluated.

Experience—my own, as well as that drawn from scientific literature—impresses me with the fact that we very seldom realize that our

'knowledge' (or, roughly, what we can *say*) *is not 'it'*, as the 'it' is always un-speakable.

Between two houses or two stones, there is some sort of sub-microscopic interaction; but on the macroscopic level, nothing definite happens. So we say that in the given context or configuration, the **units** under consideration are too heavy (implying gravitational structure), or the medium, the plenum in which they are immersed, is too light (again structural implications) and so, macroscopically, nothing obvious happens.

If that structure is changed, different conditions, different relations and results prevail. Thus, if the particles are very small, and the media not too heavy, the surface phenomena, electrical charges., begin to play an important role. We then have colloidal behaviour of enormous complexity and variability where we find, not necessarily life, but many inorganic forms, duplicating some forms found in life. Obviously, colloidal structure accounts for that.[2]

When little colloidal wholes, most probably of specialized internal structure, arise, we may have not only colloids, but also little wholes, separated by a membrane, or perhaps by surface phenomena, which represent a most generalized membrane. We may have a new structural fact, an interplay of the inside with the outside, and life begins.

The general irritability and conductivity of protoplasm is known to be strictly connected with permeability to the passage of ions and, therefore, is a structural phenomenon. On this structural foundation, physiological gradients result, forming a dynamic field of forces, again involving structure. The development of the differentiated tissue of muscles and nerves consists of higher order complex structures, based on more primitive structures; and, finally, function and behaviour of all life, man included, is due to sub-microscopic, microscopic, and macroscopic structure.

I may be reproached by specialists that, although what I have just said may possibly be true, yet, actually, to make these assertions is, perhaps, premature, in 1933, because we lack too many details.

My answer is sharp and definite, and may be considered a serious scientific suggestion, because it can be made legitimately in this form:

1) All science depends on human 'knowledge'.

2) All human 'knowledge' is structurally circular and self-reflexive, and so depends on some conscious or unconscious theory of knowledge and undefined terms.

3) Words are *not* the things we speak about; and, therefore,

4) The only possible connection between the objective and unspeakable levels and words is *structural;* introducing

5) Structural analysis of languages as fundamental; making

6) The *only possible content of 'knowledge' structural,* and

7) All science becomes a search for the unknown structure of the empirical world on all levels, and the matching of this unknown structure with the *potentially known* structure of languages; so that

8) All knowledge is hypothetical, in which

9) The most important facts must be *negative.* When the structures do not match, then we learn something quite definite about the empirical structures.

10) All predictability becomes possible because of similarity of structure; and so definitely making

11) All possible aims and quests of science uniquely structural; necessitating

12) Unique methods of translation of dynamic into static, and vice versa, in order to cover the structural exigency of both the dynamic world and the static languages.

13) Such unique methods of translation are given in the differential calculus and four-dimensional geometries, in which

14) What in a four-dimensional language is structure becomes in three-dimensional language 'importing time' function; showing once more that

15) Structural considerations are not only a modern necessity, but also the most creative and helpful for the future development of science and man, and justifying the above assertions, with the setback that

16) Full 1933 structural analysis, being one of the, or perhaps *the,* highest abstraction of this date, the mastering of that language may represent some difficulties.

The reader must be reminded (see, for details, Part VII) that the terms 'structure', 'function'., are multiordinal terms with many meanings, and so that they have no general meaning apart from context, but have definite meaning in each context. Without this realization of the multi-ordinality of terms, the statement above could not be made, for it is a structural statement about languages.

As an example of the immense and inherent importance of considerations of four-dimensional order, the following psychological experiment for which Doctor Harry Helson has suggested lately the name of Tau effect, is useful.[3]

If we stimulate three spots of the skin by touching them lightly with the end of a pencil in quick succession, and if the distance between the first and the second spots is, say, 20 mm., and that between the second and the third is 10 mm., but the 'time' interval between the second and

the third stimulations is twice as long as that between the first and the second, the distance between the second and the third spots will be judged as nearly twice as great as that between the first and second. Similar results are obtained with other analysers, such as vision and hearing. If we change the conditions of the experiment, the results may be reversed. It is interesting to note that the effect does not depend on 'knowing', as similar results happen when the subject knows the conditions of the experiment. The last shows that the experiment deals with a physiological and neurological mechanism. In general terms, if we vary the time-interval in the opposite sense from the space-interval, the latter will be distorted, showing once more the structural fact that in actual life and experience we deal *exclusively* with the four-dimensional space-time order, which, as such, must have physiological and neurological significance, and an adapting mechanism.

CHAPTER XXI

ON CONDITIONAL REFLEXES

The conditioned reflex is conventionally regarded as differing essentially from the unconditioned reflex, but this is contradicted by evidence drawn from the development of behavior. (107)

G. E. COGHILL

The main experiments of Pavlov were made on dogs, animals with a rather well-developed nervous system; and so most of what he has to say is about dogs, although some general physiological facts apply to all the higher nervous systems, man included. In some instances, because of human complexities, some results must be re-interpreted, structural linguistic re-adjustments made, and some obscuring, wrong-in-structure, *el* 'psychological' terms analysed and rejected. My linguistic, structural, *non-el, theoretical* revision leads to a new and important enlargement of the application to man of the Pavlov *experimental* theory of 'conditioned' reflexes. The fact that these independent discoveries reinforce and support each other is a striking instance of the usefulness of theoretical researches.

We must take care to notice and beware of the differences in languages. Any happening has as many aspects as there are sciences, or even human interests. Thus, if we speak about an objective 'pencil', we may speak about its chemistry, or methods of manufacturing, its uses, prices, markets,. As the content of knowledge is structural, we must search empirically for structure, understood nowadays always on three levels (the term being multiordinal), the macroscopic gross structure, the microscopic, and sub-microscopic structures.

When we deal with life phenomena, we have also different languages dealing with their different aspects. Thus, a biological language would cover eventually the vital events in general; a physiological language would be narrower and cover the analysis of phenomena in an organism, the function of its organs and the conditions and the mechanism which determine these functions; a neurological language would be physiological as applied only to the nervous system. The day is not distant when all these problems will be formulated in the language of the quantum mechanics.

A psycho-logical language is legitimate only on human levels, as we never know, or can know, what an animal 'thinks', 'feels'., and on human levels it applies to so-called 'psychic' phenomena only.

Usually, one extremely fundamental semantic fact is disregarded; namely, that what on the psycho-logical level is *objective* and in language *descriptive* to one person (e.g., 'my toothache'), is *inferential* to the other

person, and vice versa. The lack of consciousness of abstracting introduces, by structural necessity, an identification of orders of abstractions; namely, the confusion of descriptions with inferences, and vice versa. This makes it imperative to avoid psycho-logical language as much as possible. It is also bad epistemology to use a language which applies to a few individuals (psycho-logics) for describing functions which are much more general, and which, fundamentally, apply to all organisms.

It is a striking fact that, although physiology is a fairly old and well-developed discipline, yet the purely physiological approach to the study of the brain-functions is very recent, and, in the main, has been carried on by Pavlov and his followers. Pavlov gives us a simple yet profoundly true explanation; namely, that the higher nervous centres have never been treated on equal footing with other organs, or other parts of the nervous system. The activities of the hemispheres have been treated from a 'psychological' point of view, and, by analogy, we have ascribed to animals similar 'psychological' states, a remnant of primitive animism. As such attitudes have become more and more obviously absurd, we have drifted into the opposite absurdity of animalism, ascribing animal characteristics to man, forgetting that the human nervous system is far more complex, matures later than in any animal, and is a non-additive affair. Naturally, reasoning by such analogies must be fallacious.

The prevalent complete disregard of the fact that these issues are linguistic and structural makes the advances in these fields very slow and halting, and only so-called 'geniuses' are capable of breaking through these semantic barriers. Once the linguistic character of the issues is fully realized, the psycho-logical, semantic blockage is removed, freedom of analysis is inwardly established, and even 'non-geniuses' will produce important creative work. Indeed, we may find that with this realization, particularly if embodied in early education, the 'normal' man would be, what we call at present, a 'genius'.

This conclusion naturally follows if we abandon animalistic analogies and face the fact that high-grade human intelligence happens to be not less 'natural' and inherent in the history of evolution than any other 'tropism'. By eliminating the psycho-logical semantic blockage due to copying animals in our nervous reactions, we may handle this important *human function* of language properly. Man will function as *man*, in accordance with the structure of his more complex nervous system. There is no doubt at present that some organisms called 'man' have an important function connected with *s.r* called 'speech', perhaps the most complex and involved and also *unique* function evolved by this class of life, and which it does not yet know how to use. Biologically and physio-

logically, this misuse of a function must be a *non-survival tendency* for *this* class of life.

Sanity must be based on methods for the most efficient use of the human nervous system, in accordance with its structure, and will thus bring about the full working of human capacities, which at present are still semantically blocked by faulty handling of the apparatus.

Before going further, I will analyse and suggest the complete elimination from the English language of the term 'conditioned' reflex, which is structurally false to facts, and suggest in its place the uniform use of the term 'conditional' reflex, introduced by Pavlov and used occasionally by some English writers. I will also suggest the elimination of a *psychological* term, 'inhibition', from physiology and neurology, in which it should have no place at all. Such a change in language leads to new results, and also suggests new experiments. It is little known and seldom taken into consideration that long ago Locke was quite clear on the point that the misuse of language has often been taken for deep mysteries of science; but Locke, unfortunately, did not take into consideration *structure,* and *s.r*; so his arguments were, in general, non-operative.

As everything in this actual world is structurally interrelated with everything else, we should consciously look for interrelations; in which case we have to build special languages for the eventual synthesis. As we must first ascertain empirical structure, and only then coin the languages, obviously to start with a descriptive, impersonal, non-'psychological' language of ordered events on a given level is most important.

In our case, we are investigating the structural and semantic problems in connection with language. We have to accept the structural facts as discovered by physicists, physiologists, neurologists, and other scientists, and then build a language similar in structure to the empirical world. The language in which the present theory is formulated is a physiological and neurological one, as it deals with observed impersonal functionings of the organisms called 'man'. When we reach results in a physiological language, these have, naturally on the human level, a psychological aspect, and perhaps the main importance, and even value, of the present work is that it reaches the very difficult psycho-logical, semantic level by purely functional and easily controlled physiological methods.

Thus the reader must translate for himself, as nobody else can do it for him, the physiological results into his psycho-logical feelings and attitudes and evoke the un-speakable *s.r.* These must be *evoked* by the reader, otherwise he will inevitably miss the point. For instance, if it is said that 'the objective level is un-speakable', the reader should try to become entirely 'emotionally' impassive, outwardly and *inwardly* silent

about an object, or a feeling, as whatever we may say *is not 'it'*. This, obviously, involves a complete checking of affective responses, 'preconceived ideas'., making him an 'impartial observer'. In fact, to do this successfully is something very difficult to achieve, requiring long semantic training with the Structural Differential, and usually involving a complete reversal of our habitual modes of affective responses.

Similarly, when we speak of 'natural order' or reversal of this order, let the reader try actually to evoke these *s.r*, and he will find it is not so easy, as it involves a completely new process of re-evaluation. In both cases, we can gain physiological, easily operating means to re-educate the very stubborn semantic responses, by *functional and ordinal methods*. The difficulties are only serious with grown-ups; they present no difficulties in the early semantic training of infants, for whom this training becomes a powerful preventive method against future nervous disturbances (limited, of course, to this aspect of un-sanity).

Directly such a semantic re-education is accomplished, the formerly impossible is also performed, and 'human nature' has been changed. Obviously, the trouble has not been with 'human nature', but with the lack of physiological and educational means to affect the psycho-logical level and to change the *s.r*. The above applies to the so-called 'normal' man, as well as to the 'mentally' ill. It works with both types, provided the latter is in a condition to be at all accessible to approach.

The term 'nervous reflex' was originated by the mathematician Descartes. Structurally, it was a genuinely scientific notion. It implies necessity; namely, that a stimulus results in a response. Obviously, if such were not the case, an animal would not be in sufficient correspondence with its environment and could not survive. Thus an animal must be attracted and not repelled by food, it must avoid fire, and so forth.[1] The term 'reflex' is usually used in connection with A two-valued implications; this makes reflexology *el* and generally inadequate to account for organic *non-el* responses in the colloidal sense. In a $\bar{A}$-system based on the ∞-valued semantics of probability, I prefer to avoid the two-valued implication and use the term reaction instead.

The main function of the nervous system is the co-ordination of all activities of the organism for its preservation. Thus there must be no conflict between the opposing activities of different parts of the organism, and any action must ultimately benefit the whole. There may be a conflict of different excitations, but one must finally dominate the others, as otherwise co-ordination would be impossible.

If food or some noxious material is put into the mouth of a dog, a secretion of saliva is produced, either to alter the food chemically and

help digestion, or to wash the mouth out and eliminate it. But observation shows further that other factors, not food or noxious materials alone, may produce similar secretions. Thus, for instance, the sight or smell of some such material, or of the person who usually administers it, or even the latter's footsteps may produce salivary secretions.

To make the experiments as exact as possible, the dogs observed by Pavlov were submitted to minor operations. Among others, the opening of the salivary duct was transplanted to the outside of the skin, so that all secretions could be carefully and exactly collected and measured.

The principle which underlies these experiments is the observation that if we combine some hitherto neutral stimulus, such as a definite tone, colour, or shape, with the presenting of the food or acid, after a few trials this neutral stimulus acquires the potentiality of producing similar secretory effects as the food or acid itself.

This fundamental and exact method of experimentation allows considerable freedom in the selection of neutral stimuli, affecting, as we prefer, the visual, or auditory, or tactile, or other nerve centres of the animal. We can also control their numbers, intensity, their combinations, the *order* and *delays* in their application , .

If food or noxious materials are placed in the mouth, the secretion of saliva is an almost automatic reaction, owing to the physico-chemical action of these materials. Such reaction is inborn and practically general for a given species, the nervous paths for such reactions being mostly completed at birth. Not so with the reactions produced by neutral stimuli, which acquire the secretory characteristics only after some experiences. These characteristics are reactions acquired during individual life, and the nervous paths and connections have to be completed during the lifetime of the individual.

Thus, when puppies are shown meat or bread, which they had never eaten before, usually no salivary secretion appears. Only after eating meat and bread on several occasions will the sight of them produce secretions.

Some of the effects of these acquired reactions are very strong and lasting. In some experiments, the dogs were given a hypodermic injection of morphine. The usual effect of the drug is to produce nausea with profuse secretion of saliva, followed by vomiting and deep sleep. In further experimenting, it was found that the preliminaries, or even seeing the experimenter, without injection, was often enough to produce the effects of the actual injection of the drug.

Pavlov studied the nervous mechanism of the functioning of the salivary glands, not because of any special physiological importance of

these glands, but because such experimenting was the simplest, and the method used allowed him to conduct the most varied experiments under accurate control.

The experiments disclosed an amazingly subtle and complex nervous mechanism, probably typical for the functioning of other glands of internal secretions. These findings, when translated into a language applicable to the human level, disclose a great deal about the nervous mechanism underlying so-called 'associations' and other 'mental', relational, or psycho-logical semantic manifestations. Usually, the salivary glands are not supposed to be as closely connected with psycho-logical manifestations as the thyroid, the adrenals, and other glands are known to be. It is, therefore, a new and very important general discovery of Pavlov that the salivary glands have such intricate and far-reaching nervous interconnections.

The example of the dog reacting to the 'associations' (relations) of the experiment with morphine in a similar way as to the actual injection of the drug, is a close parallel to the example already given, of the patient who reproduced symptoms of hay fever at the *sight of paper roses*. In this case, the 'associations' were also uncritical, compulsory, almost automatic, of the type found in the animals. In fact, this statement is very nearly general, and we shall find later that most of 'mental' ills follow neurologically the patterns of animal responses, and so become pathological for man. This observation has very far-reaching consequences, to be explained later; but we want to emphasize it from the beginning, and to stress the fact that copying animals in our nervous reactions must be detrimental to humans.

The above narrows our problem considerably: we have to discover only the main differences between the nervous responses of animals and humans, and draw our conclusions.

The alimentary reactions to food and the mild defense-reaction to noxious materials may be roughly divided into two components, the secretory and the motor. It was found possible to link another neutral stimulus to an already acquired reaction. Thus, if a dog was trained to respond to a bell, which was a signal for food, he could be trained, further, to link a formerly neutral stimulus, let us say, the sound of a buzzer, with the bell, and the bell with food. Such a secondary acquired reaction may be called of the *second order*. Naturally, it is very instructive to find out if these responses could be extended to more orders. Experiments disclosed the important fact that, as far as dogs and *alimentary* reactions are concerned, it was impossible to go beyond the second order. However, when *defense* reactions were tested, it was found that

it was possible to establish acquired reactions of the *third order*. But it was impossible to go beyond the third order, even in these cases.

In our field, where we have to formulate sharp differences between the nervous responses of 'man' and 'animal', we say that animals stop abstracting or linking of signals on some level, while humans do not. The latter abstract in indefinitely higher orders—at least potentially.

Here we encounter a fundamental and sharp far-reaching difference between the nervous functioning of 'animal' and 'man'. This abstracting in indefinitely higher orders no doubt conditions the mechanism of what we call human 'mentality'. If we stop this abstracting anywhere, and rest content with it, we copy animals in our nervous processes, involving animalistic *s.r.* As will be shown later, this is the actual case with practically all of us, owing to our *A* education and theories. This 'copying animals' in our nervous responses is, perhaps, a natural tendency at an extremely low level of development; but as soon as we understand the physiological mechanism, we can correct our education, with corresponding human semantic results. Naturally, such 'copying animals' by humans must be a process of arrested development or regression. It must be pathological for man, no matter how severe or how mild the affliction may be. Various absolutists, and the 'mentally' ill in general, show this semantic mechanism clearly.

The reactions can be divided into two groups, those which are *inborn,* almost automatic, almost unconditional, rather few and simple, belonging to the so-called 'species'; and those which are *acquired* during individual life, allow a great variety of complications, are *conditional in different degrees,* and are acquired by the individual. Pavlov suggests different terminologies; for instance, he calls the one 'inborn', the other 'acquired'; or as usually incorrectly translated into English as 'unconditioned' and 'conditioned' respectively. The two last terms have received a scientific general acceptance, yet I would suggest that in the English incorrect translation they are *structurally unsatisfactory,* and that particularly, when applied to humans, they carry harmful implications. Structurally, 'inborn' and 'acquired' are entirely satisfactory. Terms like 'conditional' and 'unconditional' (in the original language of Pavlov), although less satisfactory, are more appropriate, as they do not imply some sort of 'cause-lessness'. In fact, the 'unconditioned' salivary reactions *are conditioned* and produced by the physico-chemical effect of the food, and so to call them 'unconditioned' is structurally erroneous. The terms 'conditional' and 'unconditional' do not have similar implications, and carry others, as, for instance, the possibility of very important

degrees of conditionality, establishing the ∞-valued character of the reactions; conditional meaning non-absolute, and non-one-valued.

For these structural reasons, I shall use the terms 'inborn' and 'acquired' or else 'unconditional' and 'conditional' reactions.

Under natural conditions, an animal, to survive, must respond not only to normal stimuli, which bring immediate harm or benefit, but also to different physical and chemical stimuli, in themselves neutral, such as waves of sound or light . , which are *signals* for animals and *symbols* for man. The number of inborn reactions is comparatively small, and, alone, they are not sufficient for the survival of higher animals in their more complex environment. Experiments have made this point quite obvious. A completely decorticated animal may retain his inborn reactions and become a kind of automatic mechanism; but all his subtler means of adjustment, owing to acquired reactions, disappear, and if unaided he can not survive. Thus, a decorticated dog will only eat when food is introduced into his mouth, and would otherwise die of starvation though food be placed all around him.

Experimental evidence seems to show that all higher activities of the nervous system, the whole signalizing apparatus, which underlies the formation and maintenance of the acquired conditional reactions, depend on the integrity of the cortex. Stimuli which produce conditional reactions are acting as signals of benefit or danger. These signals are sometimes nominally 'simple', sometimes very complex, and the structure of the nervous system is such that it can abstract, analyse, and synthetize the factors of importance for the organism, and integrate them into excitatory complexes. The analysing and synthetizing functions, as usual, overlap, and cannot be sharply divided, both functions being only aspects of the manifestation of the activity of the nervous system as-a-whole. In general, one of the most important functions of the cerebral cortex is that of reacting to innumerable stimuli of variable significance, which act as signals in animals and symbols in humans, and give means of very subtle adjustment of the organism to the environment. In psycho-logical terms, we speak of 'associations', 'selection', 'intelligence'. ; in mathematical terms, of relations, structure, order. ; in psychophysiological terms, of semantic reactions.

The language of reactions is of special interest because its structure is similar to the structure of protoplasm in general and the nervous system in particular. This language can be expanded and supplemented by the following further structural observations:

1) That reactions in animals and humans exhibit *different degrees of conditionality;*

2) That the signals and symbols may have *different orders*, indicating superimposition of stimuli;

3) That animals cannot extend their responses to signals of higher order indefinitely;

4) That humans can extend their semantic responses to higher order symbols indefinitely, and, in fact, have done so through language which is always connected with *some* response, be it only repression or some other neurotic or psychotic manifestations.

The above extension is structurally fundamental, because we can extend the vocabulary of conditional reactions to humans in all their functions. Without it, we find ourselves saddled with a vocabulary which does not correspond in structure to the well-known elementary facts concerning *human* responses to stimuli, and we relapse into the old 'behaviourism', which is structurally insufficient.

The present system is based on such observations and extensions. It was reached independently from structural and physico-mathematical considerations. With this structural verbal extension, we can easily be convinced that everything that we call 'education', 'habits', 'learning'., on all levels is building up acquired or conditional and *s.r* of *different orders*, as one of the differences between 'man' and 'animal' consists in the fact that humans can extend their symbolism and responses to indefinitely high orders, while with animals this power of abstracting and response *stops somewhere*. We establish here a sharp distinction between the high abstractions 'man' and 'animal', and so build up a psychophysiological and structurally satisfactory language.

It is obvious that the fundamental means which man possesses of extending his orders of abstractions indefinitely is conditioned, and consists in general in symbolism and, in particular, in *speech*. Words, considered as symbols for humans, provide us with endlessly flexible conditional semantic stimuli, which are just as 'real' and effective for man as any other powerful stimulus.

Take, for instance, the example of the World War! Would the men in the trenches have endured at all the horrors they had to live through if it had not been for words, and, neurologically speaking, because of the conditional *s.r* connected with words?

'If any question why we died,
Tell them, because our fathers lied.'

said the poet truly, and experience shows it is not limited to the trenches.[2]

In interpreting the experiments on animals as applied to humans, it should be remembered that some of the experiments of Pavlov, *as they stand*, would be, at the least, *neurotic* for man. The reason for this is

that the higher abstractions of man, which are due to the more developed complexities of his nervous system, would often make such simple experiments impossible. Once a conditional reaction is established with an animal, no amount of any sort of 'intellectual' persuasion, or the like, would disturb his glandular secretions, as the animal's range of 'meanings' is very limited. These secretions can be diminished or even abolished by other means, but not by 'intellectual' means alone. In the 'normal' man, his 'knowing' that the sound of the metronome or bell is part of an experiment and not a signal for actual food, would, *or should,* alter his nervous reactions and glandular secretions and make the experiments much more complex. The conditional reactions of the animals have still the *element of unconditionality.* In man, they may become *fully conditional* and depend on a much larger number of semantic factors called 'mental', 'psychic'., than we find in any animal.

On the human level, outside of the experiments with the salivary glands, we have in the psychogalvanic reaction a most subtle semantic means of experimenting with the effect of words as connected with some secretions, probably at least the sweat glands. Humans react to different events or words by minute electrical currents (among others) which can be registered by a very sensitive galvanometer and the curves photographed. It is interesting to notice that so-called 'self-consciousness' disturbs the success of the experiments, or makes them impossible, *at least with some individuals.* It should be remembered that general statements are invalidated if there are any exceptions.

In experiments, we are usually interested in their success. When analysing the ∞-valued *degrees of conditionality,* we are equally interested in their failures, which suggest a far-reaching revision of the *interpretation* of our experimental data in this field. Although some writers say that the reactions registered are 'beyond control' (unconditional), this statement, in general, is not correct, and should be amended to '*often* beyond control' (conditional of different degrees). It is impossible to go into details here, as the problems are extremely complex. In addition to this, the testing of *degrees of conditionality* presents an extremely wide *new semantic field* for experimentation which has not yet been attempted. It should be noted, however, in passing, that in these experiments different types of 'mentally' ill, as well as the 'healthy' persons, exhibit different types of curves.[3]

When psycho-logical events or *s.r* are interpreted, the difficulties become particularly acute. Thus, we seldom discriminate between the average and the 'normal' person. In the *animal world,* under natural conditions—by which is meant entirely without human interference—

the survival conditions are *two-valued* and very sharp. The animals survive or they die out. Because of this, it could be said, with regard to the animal world, with some sort of plausibility, that the average, with a long list of specifications, could be considered the 'normal' animal. We usually enlarge this notion to humans and land in fallacies, particularly in so-called 'psychological' problems, which admittedly are very difficult.

In general medical science, such mistakes are made more seldom. No physician, studying a colony of lepers or syphilitics, could conclude that a 'normal' man should be a leper or a syphilitic. He would say that probably, in a given colony, the average person is afflicted with such and such a disease, and he would keep as his medical standard for desirable health, a 'normal' man; that is to say, one free from this disease.

It is true that in the example given above, outside of such rare colonies, we have a majority which, in respect to the given disease, are healthy; so we are empirically forewarned against fallacies, although existing theories of knowledge do not forewarn us. But the main point remains true; namely, that in human life the average 1933 does *not* mean 'normal', and the standard for 'normal' will have to be established *exclusively* by scientific research. In our present work, we show that the average person copies animals in his psycho-logical and nervous processes, exhibits the unconditionality of nervous responses, confuses orders of abstractions, reverses the natural order., semantic symptoms of similar *structure* as found in obviously 'mentally' ill. Therefore, the average person 1933 *must be considered pathological.* If we take the animalistic average for 'normal', and apply it to man, we commit a similar fallacy as that of treating a colony of lepers as a 'normal' or 'healthy' group.

In conditional *s.r* of man, the average person cultivates, through inheritance and training in *A* doctrines, languages of inappropriate structure., animalistic, nervous, and so psycho-logical, *s.r.* But here, as in general medicine, the average pathological situation should not be considered 'normal'. Only a structural study can disclose what with man should be considered 'normal'. The present system performs this task to a limited degree and in various ways, among others, by the revision and the widening of the reaction vocabulary to a larger structural conditionality, as found in the, as yet, exceptional 'normal' man, and introduces the important notion of *non-elementalistic semantic reactions.*

Because of this 'average for normal' fallacy, the theories of 'conditional reflexes' in man should be thoroughly revised and enlarged to include *non-elementalistic semantic reactions;* and then we should find that often what is 'normal' with animals is quite pathological for man. The semantic difficulties are serious, because the accepted two-valued

structure of language and semantic habits reflect the primitive mythologies; so there is always the danger of drifting either into animalism, or into some other sort of equally primitive mysticism.

The net psycho-logical result of such a revision appears to be that, on structural grounds, what on human level appears as desirable, and, at present, exceptional—as, for instance, the complete conditionality of conditional and *s.r*, based on the consciousness of abstracting—ought to be considered the rule for a 'normal' man. Then the older animalistic generalizations will become invalid and reactions transformed. But for this purpose, and to be able to apply these considerations in practice, we shall have to analyse 'consciousness of abstracting' and, therefore, 'consciousness' which must be defined in simpler terms, discussed in Part VII.

When we deal with 'mentally' ill persons, the reactions which would be conditional with 'normal' persons become, in a sense, unconditional, compulsory, and semi-automatic in effect, inwardly as well as outwardly. As with animals, no amount of 'intellectual' persuasion has any effect on them, and the reactions, secretions., follow automatically. From the physiological point of view, 'mental' ills in humans compare well with *conditional reactions in animals.* It seems that under such circumstances a physiological language of different orders of abstractions, different orders of conditional and *s.r* would be structurally satisfactory. In such a language, we should pass from the inborn reactions, which exhibit the maximum of persistence, unconditionality, and almost automatic character, to the acquired or conditional reactions *in animals,* which would be called *lower order conditional reactions,* still, to some extent, automatic in their working, and, finally, to the much more flexible, variable, ∞-valued and *potentially fully conditional reactions in man,* which we will call *conditional reactions of higher orders* which include the *semantic reactions.*

In such a vocabulary, the main term 'reaction' would be retained as a structural implication; yet the *degrees of conditionality* would be established by the terms of 'lower order' or 'higher order' conditional reactions. Such a language would have the enormous advantage of being physiological and ∞-valued. Structurally, it would be in accordance with what we know from psychiatry; namely, that the 'mentally' ill exhibit arrested development or regressive tendencies.

We would say that 'mental' illness exhibits not only arrested development or regression, but we could state definitely that the *fully conditional* (∞-valued) reactions of higher order have not developed enough, or have degenerated (regression) into *less conditional* (few-valued) reactions of lower orders as found in animals. All the 'phobias',

'panics', 'compulsory actions', identifications or confusions of orders of abstractions., show a similar semantic mechanism of mis-evaluation. Although they naturally belong to the so-called 'conditional reactions', yet, being impervious to reason, they have the one-valued character of *unconditionality,* as in animals.

Similarly with the difference between signals and symbols. The signal with the animal is *less* conditional, more one-valued, 'absolute', and involves the animal in the responses which we have named conditional reactions of lower order. Symbols with the *normally developed man* (see discussion of 'normal' above) are, or should be, ∞-valued, indefinitely conditional, not automatic; the *meanings,* and, therefore, the situation as-a-whole, or the context in a given case, become paramount, and the reactions should be fully conditional—that is to say, reactions of higher order. In human regression or undevelopment, human symbols have degenerated to the value of signals effective with animals, the main difference being in the *degree* of conditionality. Absolutism as a semantic tendency in humans involves, of necessity, one- or few-valued attitudes, the lack of conditionality, and thus represents a pre-human tendency.

To what extent the language of the *degrees of conditionality* is helpful in understanding the development of *human* 'intelligence', and why a fully developed *human* 'mind' should be related with *fully conditional* reactions of higher order, can be well illustrated by an example taken quite low in the scale of life.

This example is selected only because it is simple, and illustrates an important principle very clearly. We know that fishes have a well-developed nervous system, do not possess a differentiated cerebral cortex; but experiments show that they can learn by experience. If we take a pike (or a perch) and put it in a tank in which some minnows, its natural food, are separated from it by a glass partition, the pike will dash repeatedly against the glass partition to capture the minnows. After a number of such dashes it abandons the attempt. If we then remove the partition, the pike and the minnows will freely swim together and the pike will not attempt to capture the minnows.[4]

The dash for capturing the minnows was a positive and unconditional, inborn feeding reaction, unsuited for the environmental conditions as they happened to be at that moment. The (perhaps) painful striking of the glass was a negative stimulus, which abolished the positive reaction—speaking descriptively—and established a negative conditional reaction, the result of individual experience, which, as we observe by the actions of the fish, is not flexible, not adjustable, and quite rigid,

one-valued, and semi-unconditional, or of low degree of conditionality, because, when the glass partition is removed, the pike swims freely among the minnows without adjusting itself to the new conditions and capturing the minnows.

A cat separated from a mouse by a glass partition also stops dashing against the glass, but this negative reaction is *more conditional.* In 'psychological' terms, the cat is 'more intelligent', *evaluates relations* better than the fish, and when the glass partition is removed, the cat captures the mouse almost immediately.

In this connection, an interesting experiment could be made, though I am not aware that it has been performed; namely, to separate the above fishes with a wire screen, which would be *visible* to the fishes, and repeat the experiments to test if the removal of a *visible* obstacle would alter the outcome of the experiment or the 'time' of the reactions. If the 'time' for capturing the minnows were reduced, this would mean that the conditionality of the reaction was increased, and so the seeing of the obstacle, or the increased power of abstraction, would play some role in it. Even humans are deceived by Houdinis. Are we so 'superior' to the 'poor fish'?

These problems of degrees of conditionality can also be studied in the life of insects, and the works of Professor Wm. M. Wheeler, for instance, furnish most instructive material, which we have not space to analyse here.[5]

In the process of human evolution from the lowest savage to the highest civilized man, it is natural that we should pass through a period in which the primitive doctrines and languages must be revised. The newest achievements in science indicate that the twentieth century may be such a period. Even in mathematics and physics, to say nothing of other disciplines, it is only the other day that the old elementalism and two-valued semantics were abandoned. Obviously, consciousness of abstracting produces *complete conditionality* in our conditional higher order reactions, and so must be the foundation on which a science of man, or a theory of sanity and human progress, must be built.

The suggested extension of the reaction vocabulary would allow us, at least, to apply a uniform physiological language to life, *man included.* We should have a general language for life and all activities, 'mind' included, of a structure similar to the known protoplasmic and nervous structure, not excepting the highest activities. 'Mental' ills would be considered as arrested development or regression to one-, or few-valued semantic levels; sanity would be in the other direction; namely, progression conditioned by larger and larger flexibility of conditional and seman-

tic reactions of higher order, which, through ∞-valued semantics, would help adjustment under the most complex social and economic conditions for man. The maximum of conditionality would be reached, let us repeat, through the consciousness of abstracting, which is fundamental for sanity, and is the main object of the present work, explained in Part VII.

It seems that the aggregate of inborn, almost unconditional and acquired or conditional reactions of different orders and types constitute the foundation of the nervous activities of humans and animals. The mechanism is not an additive one. A little bit of cortex 'added' involves most far-reaching differences of behaviour in life; in fact, the number of possibilities probably follow the combinations of higher order.

Higher order combinations are constructed from groups which themselves are groups. Thus, out of twenty-six letters of the English alphabet, there are probably trillions of pronounceable combinations of letters. Sentences are groups of words which are groups of letters, and their number, therefore, exceeds enormously the original trillions. Books are combinations of sentences, and, finally, libraries are combinations of books. Thus, a library is a combination of fifth order, and the number of possible different libraries is inconceivably large. As a rule, we pay little attention to combinations of higher order, disregarding the fact that even materials and the possible variety of them have some such structure.

To give an intuitive feel how combinations of higher order increase, let me quote Jevons on the simplest case, starting with 2. 'At the first step we have 2; at the next 2^2, or 4; at the third, 2^{2^2}, or 16, numbers of very moderate amount. Let the reader calculate the next term, $2^{2^{2^2}}$, and he will be surprised to find it leap up to 65,536. But at the next step he has to calculate the value of 65,536 *two's* multiplied together, and it is so great that we could not possibly compute it, the mere expression of the result requiring 19,729 places of figures. But go one step more and we pass the bounds of all reason. The sixth order of the powers of *two* becomes so great, that we could not even express the number of figures required in writing it down, without using about 19,729 figures for the purpose.'[6]

In actual life, the number of possibilities of higher order combinations are limited by structural and environmental conditions; nevertheless, the numbers of possibilities which follow such a rule increase surprisingly fast.

CHAPTER XXII

ON 'INHIBITION'

. . . "destructive lesions never cause positive effects, but induce a negative condition which permits positive symptoms to appear," has become one of the hall-marks of English neurology. (212) H. HEAD

Excitation rather than inhibition is important in correlation because from what has been said it appears that so far as known inhibition is not transmitted as such. The existence of inhibitory nervous correlation is of course a familiar fact, but in such cases the inhibitory effect is apparently produced, not by transmission of an inhibitory change, but by transmission of an excitation and the mechanism of the final inhibitory effect is still obscure. (92) CHARLES M. CHILD

But since inhibition is not a static condition but a mode of action, the mechanism of the total pattern must be regarded as participating in every local reflex. (107) G. E. COGHILL

It is highly probable that excitation and inhibition, the two functions of the nerve cell which are so intimately interwoven and which so constantly supersede each other, may, fundamentally, represent only different phases of one and the same physico-chemical process. (394) I. P. PAVLOV

The term 'unconditional reflex' applies only under 'normal' or 'natural' conditions, as we know that different drugs, such as ether, which alter the conductivity of nervous tissue., can also alter its irritability. Similarly, with conditional reactions, the introduction of *degrees* of conditionality becomes an important ∞-valued structural refinement of language, depending on, and introducing explicitly or by implication, the number of factors, the degrees of freedom., which are observed empirically, and so should have a linguistic and semantic parallel.

If we disregard, for instance, the possibility of the use of a drug., then the '*un*conditional' reactions are largely *un*conditional. The 'conditional reflexes' in animals are a much subtler form of adjustment to many more factors, and if we call them 'conditional of lower orders' we cover structurally their limited conditionality, which with higher animals is considerable. For example, a fly in the laboratory might disturb the reactions, but merely 'intellectual' interference would be ineffective. And, finally, the 'conditional reactions of higher orders' in man involve still more factors, introduce more and new complexities, and necessitate that the human reactions should be *fully conditional,* requiring ∞-valued semantics. At present, this is an exceptional occurrence, although the potentiality for such *full* conditionality is present in the majority of us.

The mechanism of the unconditional reaction is, under ordinary circumstances, almost automatic. It is evolved on the background of

general protoplasmic characteristics, combined with structural polarity, symmetry., and is not efficient enough for the survival of higher organisms.

Under more complex conditions, the adjustment for survival must be more flexible: a similar direct stimulation must, under different conditions, result in different reactions, or different stimuli, under other conditions, produce similar reactions, resulting, ultimately, not only in direct responses to stimuli, but also in the equally important holding in abeyance of the reaction, or even the abolishing of it. Let us assume that the direct response of a cat to a mouse would be clawing and chewing. If that given cat would just claw and chew when the mouse was some distance away, I am afraid such a cat would soon starve, for such an immediate response would not be a survival response, and this characteristic could not become hereditary. The cats which have survived and perpetuated their characteristics are, as a rule, different. When they see, hear, or smell the mouse at a distance, they flatten out, keep still, crouch., and get ready, until they are in such a position that a jump will procure the victim, and not merely frighten it away.

We see that, under more complex conditions, the nervous mechanism must produce not only direct responses to the stimuli but also equally important delays and temporal or permanent abolishments of these direct responses to stimuli.

Hitherto, we have analysed the simplest reactions of a positive character in which a stimulus produces a direct and obvious response; e.g., the showing of the food or the ringing of the bell results in an excitation in the nervous system and the secretion from the salivary glands. We are, however, acquainted with another type of fundamental nervous activity of equal importance. For instance, in experimenting with the positive reactions, we must be careful not to introduce any extra stimuli, as any new stimulus immediately excites an investigatory reaction, and the alimentary conditional reaction becomes temporarily abolished. From our personal experience, we know a large number of stimuli which have some such hindering effect on our respiration, circulation, locomotion., which we describe as 'paralysed with fear', 'speechless with rage', 'struck dumb', 'stupefied with pain',. The diminution, or deviation, or the lack of some function or response on the nervous level is usually called 'inhibition'.

The term 'inhibition' is structurally a profoundly unsatisfactory and a misleading psycho-logical term, and should be *completely* abandoned in physiology and neurology, although it could be retained in psycho-logics and psychiatry. This term is in general use, and the sug-

gestion of abandoning a term in general use is always hard to accept. Therefore, it will be well to analyse it in some detail. In this case, it does not matter if the positive suggestion of a new term or terms is structurally acceptable; the analysis of the term 'inhibition' shows clearly that it has false to fact implications, and so should be rejected in neurology in any case.

This term is a favorite word in ecclesiastical and legal literature, and means, in the main, to forbid, to prohibit, to hinder, to restrain. It is a psycho-logical term; it implies anthropomorphic 'free will' and 'authority' notions perfectly unfit for *neurological* use. It is not an exaggeration to say that the structural implications of this term underlie the older animalistic prohibitive and punitive education, legal and ecclesiastical tendencies, which, in 1933, are known to be not only in a larger sense inoperative, but positively harmful. On the human level, this word is, perhaps, responsible for the fact that so much about our educational and social methods is uncertain and often harmful. Education is a process of building conditional and *s.r* of different orders. If the *neurological* terms dealing with conditional reactions are structurally unsatisfactory, our speculations which are carried on in these terms must involve these false implications. When the empirical results are unsatisfactory, as they must be, because of wrong structure of the arguments, and a scrutiny of our argumentation shows them to be correctly following the structural implications of the language used, then we usually blame 'human nature', which is a very unintelligent excuse, indeed.

The implications of the term 'inhibition' become a guidance for our conduct; we repress and, in consequence, breed un-sanity and maladjustment. On animal level, 'repression' is workable, but, on human levels, we need a subtler regulative mechanism, in accordance with the structure of the *human* nervous system, and this is found in the fuller conditionality of reactions, based upon consciousness of abstracting, and involving, of course, affective components, semantic factors of evaluation which regulate human impulses without the animalistic repression. In humans, the 'inhibited', repressed impulses *often remain as internal excitatory factors;* they are not eliminated by some 'supernatural' hocus-pocus, but remain active, sometimes very active, semantic sources of internal excitation, resulting in conflicts which generally have pathological results.

We are usually told that 'inhibition' plays an important role in conditional reactions. With the introduction of the *degrees* of conditionality, the importance of the possibility of altering, delaying, or abolishing some immediate response becomes much more accentuated. Indeed, it appears that this possibility of influencing responses is an important factor in the

mechanism of conditionality of lower orders, but becomes the *main factor* in establishing the *degrees* of conditionality of higher orders. Obviously, the reactions become very labile, the adjustment to conditions very subtle, allowing the organism to survive under the most complex conditions, such as are found in highly 'civilized' life.

This mechanism is responsible not only for human intelligence, but also for all that is constructive in so-called 'civilization'. Vice versa, for survival in such complex civilization, one must possess these *fully* conditional reactions. At this point, it will suffice to mention that in organisms below humans, 'inhibition', which underlies the mechanism of conditionality of reactions, plays a most important biological and survival role, while on the human level it is the foundation on which all human *s.r*, 'intelligence', and desirable human characteristics are built. The present theory introduces methods to make the application of the above considerations possible in daily life.

All possible analysis depends not only on definitions of terms but also on *undefined terms,* which, outside of mathematics, have seldom, if ever, been investigated, thus making the structural assumptions which they introduce unconscious. In definitions, we also usually posit structure, though we seldom realize this fact. When we approach the experimental side of science, which is the search for empirical structure, the implications involved consciously or unconsciously in the defined and undefined terms play a very important role, and they direct, to a large extent, our efforts and ingenuity. This is why we still have so few genuinely creative scientists, although since the psycho-logically and semantically liberating work of Einstein, the number of creative physicists of the younger generation has increased surprisingly. Yet the majority of scientists do not realize to what extent their *s.r* are influenced by the terms they use and what enormous help and creative freedom they would have from being conscious of the role the structure of language plays.

With this realization, before we begin the constructive analysis of such an important term as 'inhibition', we must state clearly what the general biological presuppositions which underlie such an analysis are.

The present work is a $\bar{A}$-system, structurally very different from the older systems, which attempts to build a verbal system of similar structure to the empirical structures, as given by science 1933. The older systems had also a structure similar to the very limited knowledge of empirical structure which our primitive ancestors had. Hence animism, anthropomorphism, 'psychologism', and the rest, and the per-

sistence of such structural features in science as 'inhibition' in neurology, 'force' and 'heat' in physics..

According to scientific standards of 1933, there is, as far as I know, only one biological system in existence which can be called modern, and this is the $\bar{A}$ biology of Professor C. M. Child (see Chapter VIII). It is, therefore, necessary to accept this system, and also the $\bar{A}$ neurology of Professor Herrick, which is based on this biology.

Generally, the neurologists tell us that the structural aspects of 'inhibition' are unknown. To a large extent, this is true, although it is quite obvious that a *'psychological'* term cannot shed any light on its physiological structure. To get glimpses of this mechanism, we must start our analysis quite low in the scale of life and see what the most general characteristics of protoplasm are.

All protoplasm is irritable. In any undifferentiated bit of protoplasm an excitation must (1933) spread in a diminishing gradient, establishing, by necessity, a region of highest excitation in contact with the stimulus, resulting in a polar orientation, with an eventual future head-end, and establishing a physiological gradient, long preceding the appearance of any differentiated tissue. The nervous system is a later outgrowth of such an oriented dynamic field, and its primary morphological and physiological characteristics are, to that extent, predetermined, being, in the meantime, a joint phenomenon of the inherent characteristics of protoplasm, its irritability, conductivity, and what not, and of its reaction to the environment. The physiological gradient is, then, the simplest and the most general primary reaction arc in a given individual, and constitutes the physiological basis for the structural and functional development of all other arcs.[1]

Amoebas are primitive little aquatic animals of approximately spherical symmetry which have no differentiated organs at all; yet they show quite complex reactions and various adaptive activities to be found in higher animals. The amoebas can pursue their victim, show preference for stimuli, and move away from the prick of a pin, select their food,. This fact shows that protoplasm, so little differentiated, and, from the organic point of view, undifferentiated, exhibits both muscular as well as neural characteristics. This fact is fundamental. It shows that in colloids which happen to be sensitive and which possess a special type of conductivity, which, from a physico-chemical point of view, is only a special aspect of one mechanism, there is already present the potentiality for any further development. Professor Child's physiological gradients, the structural precursors of the nervous system, are a necessity, because of the dynamic potentialities of the plenum and the necessary relation

to the environment, as there is no such thing as anything without environment. The stimulus, in the meantime, establishes structurally a functional polarity as a fundamental characteristic of all, even most primitive, protoplasm, and as the result of the contact of sensitive and conducting colloidal structures with the environment.[2]

In sponges, which have primitive muscular tissue but no nervous system, the muscular tissue exhibits also both characteristics, combining receptive and motor functions, showing that from the start the supposed muscles are, in reality, neuro-motor organs.[3] The actinians have no central nervous system. By the aid of an incision, we may produce in them special additional growths of tentacles sometimes with a mouth, sometimes without. If, in the last case, we place a piece of food in the tentacles, they will bend toward where the mouth should be. If we cut such a tentacle away from the body, we still find that in contact with food it will bend in the one direction. But here we are dealing not only with the sub-microscopic dynamic structure but with macroscopic structure, where the irritability and the structure of the peripheral organs determine the reaction.[4]

When we experiment with animals with a more developed nervous system, such as ascidians or worms, we come to new and very instructive facts. Loeb has removed the ganglion from a number of *Ciona intestinalis,* a large transparent ascidian, which normally, when touched at the oral or aboral opening, closes the openings, and the whole animal contracts into a small ball. It appears that a few hours after the operation mentioned they relax. If a drop of water falls on such an animal, the characteristic reaction appears again, showing that the reaction was not due to the ganglion but is determined by the structure and arrangement of the peripheral parts and the muscles. The nerves and the ganglion play only the main role as a quicker conductor for the stimulus.

Even in higher animals we find vestiges of such primitive generalized mechanisms. For instance, Loeb, in his experiments in removing the brain from sharks, found that, even after death and when signs of decomposition had already begun, light produced a contraction of the pupils.[5]

In a decapitated worm, practically all normal reactions are retained. If we cut the nervous system of a worm in two, the two parts of the worm move in a co-ordinated way as long as they are connected by a little bit of tissue. The experiments were carried further: a worm was cut in two completely, the two halves were connected by a string, and they still moved in a co-ordinated way, showing once more that originally the nervous system was a specialization of general protoplasmic characteristics of irritability and conductivity and structure, which, at present,

are known to be strictly interconnected.[6] Multiordinal structure is the explanation of this behaviour. Similar examples could be given in great numbers, all of which would support the above well-established view.

Among the general protoplasmic characteristics we do not find 'inhibition', but only positive excitation and conductivity. This issue is fundamental and should be taken as a foundation for further analysis.

If a wandering amoeba comes to an illuminated spot, the animal will not remain in that region. Here is, seemingly, a new fact, and we must *select* the language we want to use in this connection. If we follow the old animism and anthropomorphism, we could say the animal 'knows'., or that some 'demon' has forewarned it, or, with equal justification, say that it is an example of 'internal inhibition' or 'prohibition'. The introduction of such terms, of course, explains nothing physiologically, but simply multiplies metaphysical identifications on the unconscious yet false to fact assumption that a word 'is' the thing we are talking about—a vestige of the primitive 'magic of words'.

Loeb pointed out long ago that to be forced to introduce animism and anthropomorphism is enough *to neglect the analysis of an external stimulus*. This is true not only in biology, physiology, neurology., but also in physics. The difference between the N and $\bar{N}$ systems depends on the fact that Newton did not take into consideration the character of the stimulus, the finite velocity of the ray of light, which is fundamental in any observation, but that Einstein did take this into consideration. The ∞-valued determinism (the restricted principle of uncertainty) in the newer quantum mechanics depends on taking into account the disturbing effects an 'observation' has on the 'observed',.

What are the known facts in the meantime? Let us start with the character of the stimulus, light. We know, positively, that light can be considered a very potent stimulus, and so the behaviour of the amoeba was a direct response to this stimulus. In fact, we know a little about this mechanism without introducing any 'demons' or 'internal inhibition'.

The starfish of a certain species has a symmetrical structure consisting of five arms. Its nervous system consists of a central ring around the mouth and peripheral nerves radiating from the ring into the arms. If such an animal is laid upon its back, it will right itself, but it is essential that not all arms should move simultaneously. In a normal animal, having five arms, usually three arms do the work and two of them remain quiet. If we destroy the nervous connection between the arms, this co-ordination is destroyed; all five arms begin to struggle, and the starfish cannot right itself, unless by accident. Should we again invoke 'demons' or 'inhibitions', or analyse the stimulus-complex and its effect?

Obviously, when the starfish is put on its back, a new stimulus-complex is operating upon it, resulting in a complex adjustment.[7]

As we already know, any stimulus applied to a bit of living protoplasm, because of the colloidal structure and of the inherent irritability and conductivity of the plenum, produces a physiological gradient, establishing, thus, some sort of polarity, symmetry, relations, order, and structure, and indicating what structure our language should have. Again, no trace of any 'inhibition' or 'prohibition' is found, and on the silent, un-speakable, objective level everything happens the usual way, without any regard to, or respect for, our *talking*. Talking only becomes a very genuine danger when on language of primitive structure we build our creeds, institutions, rules of conduct . , and our methods of investigation. In the last case, our sciences are nearly as slow, halting, perplexing, difficult, non-co-ordinated, and, in a larger sense, ineffective, as our creeds and institutions have proven to be. Our sciences may have added to our comfort, but, outside of psychiatry, they have not contributed much to human happiness.

As structure seems so fundamental and can be discovered everywhere, we should not be surprised to find that in structure, or perhaps, still better, in the general structuro-sensitive-conductive dynamic complex with definite structure on different levels, we shall find the solution for obvious positive reactions of organisms, as well as for the lack of them.

It is not possible or necessary to go into further details here. The structural data, however, although they are not particularly emphasized, are given in handbooks of physics, colloidal chemistry, chemistry, biophysics, biochemistry, biology, physiology, neurology , . At present, it is realized in science that structure is of extreme importance ; but, because of identification, it is not realized that structure is *the* only possible *content* of science and of all human 'knowledge'. This fact, of course, makes the quest of science uniquely structural. Because of it, we come to a very far-reaching general rule, that all 'understanding', to be such, must exhibit or assume structure, thus formulating the supreme aim, and, perhaps, uniquely indicating the only possible method, of science.

Two more simple examples may be helpful. *Mnemiopsis* or *Eucharis* have swimming plates which beat rhythmically, with considerable regularity. When the plates are stimulated mechanically, the movement ceases in the presence of sufficient calcium salts in the water. In similar media, but containing no calcium, a mechanical stimulus does not stop the movement of the plates, but just the opposite. It accelerates their motion, showing clearly that the effect of direct stimulation can

be reversed when the structural relations are altered. Once more, no 'demons' and no 'inhibition'.[8]

In higher animals, we usually find a well-developed symmetry and muscles of which the activities oppose the results of the activity of other muscles. Such muscles are called antagonists. If two antagonists of equal strength are stimulated equally, no macroscopic effect of the stimulation of both muscles results. If one of the antagonists is stronger than the other, the macroscopic effect of the stimulation of both muscles results not in some general convulsion, but in a one-sided action of the stronger muscle. Obviously, these results are the necessary consequence of structure on different levels. We had, in the first case, a lack of obvious macroscopic reaction, although stimulation was present and did its work. This was due to structure.

It is known that some drugs, such as strychnine or the toxin produced by the tetanus bacillus, produce a state of *general* and high irritability of the nervous system. The slightest stimulus to the surface will produce a spasm which affects practically every muscle of the body. The pinching of the foot, instead of producing a withdrawal, results in the rigid extension of the legs, arms, and back. The extension is no longer a co-ordinated process, but is associated with strong contraction of the flexors, the final state of the limbs being determined by the surpassing strength of the stimulated extensors. The effect of the tetanus toxin is similar. In a monkey, under normal conditions, the electrical stimulus of a certain spot of the cortex will produce the opening of the mouth; similar stimulation of another spot will produce the closing of the jaws. But, under the effect of the toxin, the stimulation of *any* of these spots will produce the *closure* of the jaws, because any attempt to open the mouth will excite the stronger masseter muscles and effectively close the mouth.[9]

The above examples show again that no 'demon' or 'inhibition' has prohibited the withdrawal of the foot or opening of the mouth, but that the excitation of stronger antagonists is responsible for the result—or, if we wish, for the lack of results. All of which is obviously structural.

All the above discussions and examples—and they could be expanded and extended to fill volumes—show clearly: (1) That in the structurally more complex organisms the process of co-ordination and adjustment to more and more complex environmental conditions, leading to wider activities and fuller conditionality of reactions, is partially based —to the extent of one-half, or even more—on the lack of direct response to a stimulus, leading to delayed action and involving the four-dimensional order, all of this being a function of the entirely general charac-

teristics of protoplasm; namely, its structure, excitability, and conductivity (the last two characteristics being also a result of sub-microscopic structure) without the intervention of 'demons' or of 'inhibition'; (2) that in every case there is an *excitation,* no matter whether the result is a positive or a negative reaction, or whether we can, at present, trace it in detail.

As Professor Herrick says: 'On this view of the situation the supposed inhibitory effect of the cerebral cortex resolves itself into a differential dynamogenic cortical influence. This is partly specific and phasic, acting upon particular subcortical functional systems while these are in process and tending to depress all conflicting activities either by withdrawing available nervous energy from their apparatus of control or by equal activation of agonist and antagonist systems with resulting stasis. It is partly a general and tonic activation or reinforcement of all lower reflex systems. Upon removal of the visual cortex the specific phasic activation of learned reactions is abolished. Upon removal of the entire cortex the general tonic cortical effect is abolished. The operation has not stimulated inhibitory fibers, as some have supposed; it has removed the sources of tonic activation which normally are always operating.'[10]

'The cerebral cortex from its inception exerts more or less inhibitory influence upon subcortical functions. In the simpler learning processes of rats there seems to be a differential activation of some key factor of a subcortical learning process . . . which in effect draws off all available cortical energy, leaving other and irrelevant sensori-motor processes relatively enfeebled so that they are subordinated. The effect is the same as if a specific inhibitory action were exerted by the cortex upon the inappropriate movements . . . It may be suggested, further, that all inhibition is in reality a differential activation, the mechanism being in some cases simply the "drainage" phenomenon . . . and in other cases this effect supplemented by positive activation of two antagonistic motor mechanisms so that their interference blocks all reactions of non-adaptive sorts.'[11]

In these statements of Professor Herrick, we find a language of similar structure to the known facts. The terms of *differential dynamogenic cortical influence* and *differential activation* cover all known facts, and may cover future facts, because the terms are structurally very flexible, and will always allow us to enlarge our knowledge of the mechanism of the so fundamental *differential activation.*

The difficulty in eliminating the term 'inhibition' and suggesting a new physiological term to take its place is considerable, because this term is used in many different forms and meanings. The term 'inhibit' is

used in its various forms as a substantive, an adjective, a verb, an adverb, sometimes as a psycho-logical term, sometimes as a physiological one, yet *never* carrying physiological implications, but always psychological and anthropomorphic ones connected with its origin and standard use. It was introduced into science when physiology and neurology were in their infancy, and so were still under the influence of primitive animism and anthropomorphism.

The term, because of its character, is not scientifically descriptive. It does not suggest functional, actional, directional, or other structural implications, but suggests notions irrelevant to science connected with its origin and standard use, making it a far-fetched inferential term, the use of which must retard the advances of these sciences.

Once we introduce a physiological term with physiological and, therefore, structural implications, our expressions will have to be reshaped to make the use of the term possible. Such re-wording will always carry quite definite structural implications, which, in turn, suggest further experiments in the search for structure and so have a *creative* character, not to be disregarded. Thus, as we have already seen, the term of 'degrees of conditionality' suggested further experiments and the revision of older data.

This statement is quite general and may be summed up as follows: The introduction of a new structural term may: (1) eliminate the improper implications of the older terms; (2) introduce new and creative implications which suggest the need of verification and so lead to new experiments.

At this point, I suggest a term which may be useful and will, perhaps, be acceptable for scientific use. As the fundamental character of 'inhibition' seems to be 'differential activation', the term to be coined should possess two main structural implications: (1) it should be directional, or indicate the sense of the reaction, and (2) it should imply activation.

We find such a term in 'negative excitation', 'negative stimulation', 'negative activation', 'negative phase'. , and it is possible to extend the use of this term by making as many compound terms as we need.

If possible, we should have terms which help us to keep on one level of analysis, and so automatically prevent us from confusing levels, since modern science always deals, at least in principle, with not less than three levels, the macroscopic, the microscopic, and the sub-microscopic, thus making confusion quite easy. If we call the positive effect of a stimulation on the macroscopic level 'positive', any other stimulation which might fail to produce the positive effect on *this level,* or which

might counteract it, would be negative. The implication would remain that there was some excitation, but that it did not produce the effect which we had called positive. Structurally, such a term would be satisfactory, especially as it would help us to keep on one level of analysis and not confuse the main levels through verbal structure.

Such a language would help us to study the mechanism of 'differential activation', and would carry helpful implications. If any cases appeared in which this term did not cover the field, either the term could be enlarged, keeping the implications, or the statements should be altered so as to be expressible by such terms. The last would always prove to carry interesting implications, suggesting experiments.

In the processes going on in the nervous system, there is no occasion for the application of terms like 'prohibition' or 'inhibition'. There is no standstill in these sub-microscopic processes, though the manifestation on the macroscopic levels can be either of a positive or of a negative character. On the sub-microscopic levels, there is a nervous excitation which often stimulates antagonistic processes., with results which are not always obvious.

The implication of the term 'negative excitation', although limited, is structurally correct in 1933. Without going into full detail here, I merely suggest a few considerations. First of all, the term preserves its main implication; namely, that of excitation, 'negative' suggesting that this excitation takes an opposite course to the positive one. If, for instance, a positive excitation produces, let us say, the activities of the salivary glands, a negative excitation in this respect will not produce them but will produce other activities, such as, for instance, an investigatory reaction. With a negative excitation, there is an excitation, but it produces different results. There is no possibility of stopping or prohibiting or inhibiting nervous activities, short of death as-a-whole or destruction in parts; but only a possible deviation of activities, owing to enormous possibilities in establishing nervous connections, endlessly subtle dynamogenic effects,.

In some instances, 'inhibition' might be regarded as a form of nervous exhaustion; but such a notion cannot always be structurally correct, as there is much evidence at hand that 'inhibition' spreads to other cortical elements which were not functionally exhausted, or that it can be counteracted by some new excitation. 'Inhibition' thus preserves its *active* character. The origin of 'inhibition' is also very instructive, and a mass of experimental data shows that it can be produced experimentally. Among other ways, it can be produced by very weak, very strong, or unusual stimuli, *but stimuli, anyway*. As a rule, any

extra *nervous excitation* in the central nervous system manifests itself at once, either in diminishing, or in completely abolishing (temporarily, at least) the conditional reflexes prevailing at the date.[12] If we find that exhaustion is, in some instances, the structurally correct term, there is no reason why we should not use it, instead of using a *psycho-logical* term of 'inhibition', on neurological levels.

That the terminology of positive and negative *excitation* is structurally appropriate finds its further support in the so-called 'disinhibition'. Thus, an 'inhibition' of an 'inhibition' reverses the neural process prevailing at a given 'time' and becomes a positive excitatory one. In our language, because of structural considerations, we should say that 'disinhibition' should be labelled as 'negative excitation of *second degree*', resulting in a positive excitation. If we were to 'inhibit' 'disinhibition', we should have, again, 'inhibition',. With the new terminology, it would be a negative excitation of the third degree, which would give negative results, and a general rule could be established, in complete accordance with the mathematical language in which the even degrees of a negative excitation would have positive characteristics and the uneven would remain negative ('inhibitory').

Such a language would not just borrow 'by analogy' some mathematical features. Once we take structure into consideration,—and linguistic issues represent an adjustment of structure—when a systematic analogy is found, it has always structural implications which should be used for testing structure. There can be no serious objection to the statement that mathematics is, at present, a limited language of which the structure in 1933 is similar, or the most similar we have, to the known structure of the world and our nervous system. The use of such language must be always desirable, as it is a test of structure and so leads to further discoveries of the unknown structure of this world. To the best of my knowledge, the above is a novel, very general, *structural* use of mathematics considered as a prototype of languages. Our emphasis is now on the *structure* of mathematics, and not on the numerical solutions of equations, the *possibility and usefulness* of which is precisely due to the fact that equations express relatedness, and so necessarily give us structural glimpses.

From a structural and linguistic point of view, the historical development of mathematics shows that it is a first successful attempt to develop a language with a structure similar to the empirical structures, and shows the ideal conditions of producing languages.

When we had only positive numbers, we could add two and three and make five, we could subtract two from three and have the remainder

one, but we could not subtract three from two. Yet the structure of this world is such that a further development in the structure of the language was imperative. Thus, if an object moves in a given direction with the velocity two feet per second, and some external factor imparts to it a velocity of three feet per second in the opposite direction, the original direction of motion will be reversed, and the object will move with the velocity of one foot per second in the opposite direction. Or, to give another example, some one has two units of money and he buys something which costs three units of money. He is then in debt one unit.

Such facts necessitated the introduction of negative numbers and so made subtraction always possible. If the motion in one direction or the amount of money in our pocket was called 'plus two' units, and we subtract from it three units, the results were 'minus one', meaning a conventional reversal of direction, or sense, for motion, or a debt, instead of a possession, for money.

Experimental facts of division again necessitated the expansion of this language. Thus, fractions were introduced so as always to allow of linguistic division. The 'imaginary' number, $i = \sqrt{-1}$, was introduced to allow, in all cases, the extraction of roots , . For a long 'time', the number $i = \sqrt{-1}$ was considered almost mystical, but, of late, when a physicist or an engineer finds it in his equations, it is almost an unmistakable indication for him to look for some wave-motion in the world. More extended observation of the empirical world and structure required further structural adjustment of our languages.

In the vector calculus we have the so-called scalar product which obeys the ordinary laws of multiplication and $a.b = b.a$ where the order of the factors is of no importance. The vector product does not follow these rules, as the order becomes important; thus, in a vector product, $a.b = -b.a$. In the newer quantum mechanics, to account structurally for the experiments, still newer numbers were introduced. Instead of the old arithmetical $qp = pq$ or $qp - pq = 0$, we introduce new numbers where $qp - pq = \frac{ih}{2\pi}.1$.

It is very significant that a similar linguistic evolution appears justifiable in the case of the function of the nervous system in general and in the structure and function of the conditional reactions in particular. As experience and theory show, the fundamental structures and functions we find in life are not 'plus' affairs, but represent some higher-degree functions of a non-additive character. The typical functioning of the human nervous system (time-binding) is represented by an exponential function of 'time'.[13] Now we see that the reversal of the sign

of negative excitation also follows exponential rules, and experiments show that the change in order of abstractions which, by necessity, must be passing from even to uneven numbers of orders or vice versa, also reverses the sign of the reaction (see Part VII).

In the case of positive excitation, there is also a structural parallel with the newer languages of mathematics, but we do not need to analyse it here, because the foundation of the more flexible, adjustable responses begins with a negative effect; and, in this case, the language I suggest is fully justified without further explanations. The neurological importance of 'consciousness of abstracting' is based precisely on the fact that it automatically involves a fraction of a second of psycho-logical delay, and thus is fundamentally based on, and introduces in training, a wholesome 'inhibition'.

We come thus to a weighty structural conclusion that the fundamental processes of the nervous system are not only non-plus processes but that they follow the exponential rules of signs. As soon as we realize that from a structural point of view 'structure' and 'function' are only different types of language in which to speak about two aspects of what is going on, on the silent un-speakable level, and that on this level these two aspects *can never be divided,* we must also build a *non-el* language. Such a language is found in *dynamic structure,* out of which arises function, and even macroscopically relatively enduring structures as special aspects, and the exponential character of the fundamental activities of the nervous system becomes a necessity.

In modern mathematics numbers can be interpreted as operators., which, in our case, suggests great freedom of structural use, and widens the application of these notions.

To put the problems as simply as possible: all the more subtle forms of adjustment in organisms, 'intelligence', so-called 'civilization', our 'ethics', 'happiness'. , and, finally, *sanity,* which is the evidence of semantic adjustment or proper evaluation on human levels, are based on the neurological interplay, the number, and multi-dimensional order of superimposed (not added) positive and negative excitations. The positive, or the direct and obvious, responses are the more primitive; the negative, resulting in not always obvious consequences, are the result of further structural complexities, which reach their culmination in the normally developed highly cultured man.

Such indefinitely superimposed negative excitations are found physiologically in the hierarchy of higher and higher orders of abstractions, which are able to reverse the sign of the *s.r,* and so, structurally, make these considerations extremely workable and neurologically sound,

and justify their introduction and use. This accounts for the fact that what was evaluated as tragic or painful, or joyful, or shameful., to one generation or culture, does not seem so to another. Our personal difficulty usually is that, at present, we copy animals in the relative unconditionality of our responses, because we are not acquainted with this semantic mechanism. We are not prepared to change in one single generation the sign from a minus to a plus, or vice versa, without a great amount of struggle and semantic discomfort.

Now, such discomforts are usually harmful to the human nervous system, but the structural understanding of this mechanism helps us to eliminate these semantic pains, and so leads toward nervous balance and sanity.

It seems that the neurological mechanism operating in this connection is similar to the one formulated by Pavlov, thus: 'Two facts relating to the central nervous activities stand out clearly. The first is that the extraneous stimulus acting on the positive phase of the reflex inhibits, and acting on the negative phase dis-inhibits, in either case, therefore, reversing the nervous process prevailing at the time. The second is that the inhibitory process is more labile and more easily affected than the excitatory process, being influenced by stimuli of much weaker physiological strength.'[14]

Negative reactions or 'inhibition' must be interpreted as the neurological foundation of 'human mentality', and the result of external and internal stimulations. Because of structural interrelations, the main factor of building human 'mentality' and developing internal 'inhibition' must be more labile and must be influenced by stimuli of much weaker physiological strength.

This explains also why the solution of our problems in education, social life., must be not the animalistic external 'inhibition' alone, but must become, in the main, special internal 'inhibition', effective and yet harmless to the individual nervous system. All of us possess this most general nervous mechanism. The problem is to discover the means to operate it. We shall see later that in consciousness of abstracting we find a workable semantic solution, allowing an automatic change of sign of the reaction. It should be recalled here that all stimuli and all responses are complex, the word 'simple' being structurally false to facts. On the human, and particularly on the linguistic level, it is practically never possible to ascertain an 'absolute' order of abstraction, or the degree or order of an excitation. These are often the results of racial time-binding, and extremely complex, nervous processes, and every superimposition of a new neurological process (not addition) may fun-

damentally alter the whole character of the *s.r* and reverse the sign. In negative excitations, the passing from one degree to another changes the sign of the reaction. In practice, we are only interested in two neighbouring levels of abstractions or in two neighbouring degrees of negative excitation, simply because these involve, by necessity, a passing from an even to an uneven degree or vice versa—in both cases reversing the sign of the *s.r.*

The general organismal adjusting mechanism of the 'investigatory reaction' responds positively to a new stimulus, but with very important survival value acts negatively on established positive conditional reactions in animals. It is, at present, much weakened and often ineffective with man, resulting in non-survival, non-adjustment and 'mental' ills for man. It is a well-established fact that different stimuli either interfere with each other, resulting in modified behaviour, or reinforce each other and have cumulative effects. On the human level, different 'mental' factors play the role of internal positive or negative excitatory semantic complexes, which, because of verbal conditions (and all doctrines are *always* connected with an *affective* background), may reinforce a given stimulus, thus making its physiological effect variable and of different strength. Under such conditions a new stimulus does not produce the investigatory reaction with all its beneficial results. This mechanism is, perhaps, responsible for the well-known fact that primary instincts with humans are, by far, weaker and more variable than with animals; whence it comes that *humans seldom know by themselves, without science, what is best for them.*

We should not be surprised to find that under these more complex conditions human investigatory reactions may be of different types, culminating in the *typically human* investigatory reaction, which would introduce the natural, yet more important, *delay in an immediate reaction* to a former stimulus. We shall find, later, that consciousness of abstracting is such a distinctly human and very useful investigatory reaction that on the human complex semantic level brings relatively as much benefit to the human organism as it does on the animal level to animals.

It seems that the nervous mechanisms of both types are similar, except for the fact that on the human levels we have more factors which are external and internal stimuli than on the animal level. If we copy animals in our nervous processes, we are, in reality, worse off than the animals, because, with our more complicated nervous system, it means for us a pathological condition.

CHAPTER XXIII

ON CONDITIONAL REACTIONS OF HIGHER ORDERS AND PSYCHIATRY

In the dog two conditions were found to produce pathological disturbances by functional interference, namely, an unusually acute clashing of the excitatory and inhibitory processes, and the influence of strong and extraordinary stimuli. In man precisely similar conditions constitute the usual causes of nervous and psychic disturbances. (394) I. P. PAVLOV

The fact that the maximum disturbance in the central nervous activity does not appear immediately on administration of the causative stimulus, but after one or more days has been observed in many animals. (394)

I. P. PAVLOV

Psychiatrists will readily understand the structurally false to facts and harmful implications of the term 'inhibition' on the *neurological* level, when they consider that often 'pain', 'fright', and different 'prohibitions' and 'inhibitions' on the psycho-logical level result in nervous processes which are not *passive, eliminated factors,* but remain what they were originally—exciting semantic factors 'repressed' on human levels—and become very active and potent causative factors in many 'mental' and physical ailments.

If the *non-el* point of view and language are seriously applied, there seems to be no escape from the conclusion that the future physician, on perfectly scientific, structural, physico-chemical, and colloidal grounds, will never attempt to divide the 'physical' from the 'mental', and different nervous processes now called 'inhibition' will come prominently to the fore as active, to be taken care of and *never* to be disregarded.

That the mechanism of conditional reactions in animals bears an astonishing resemblance to the mechanism of 'mental' ills in humans, because of the relative unconditionality of both, is exemplified practically throughout the whole work of Pavlov, although he did not point out this particular connection. As soon as this is understood, we shall find that some of the experiments of Doctor Zavadzki, made in Pavlov's laboratory *twenty-five years ago,* disclose a neurological mechanism which underlies practically all psychotherapy, and which, therefore, appears very important and to deserve special discussion.

I do not know the percentage of the successful application of psychotherapy of any scientific school, or of extra-medical cults, because the many cases of failure are very seldom recorded. We usually forget, or do not realize, that the successful cases teach us, structurally, *less* than the failures, because there is always an infinity of ways in which we can account for a positive result, which is structurally entirely invali-

dated as such by a single failure, if the possibility of it is not foreseen by the structural flexibility of the general method.

From what I gather (though I may be mistaken) of every hundred patients who seek relief in psychotherapy, fifty fail completely. The remaining fifty can, perhaps, be divided into two groups; the first one of, say, ten, who become entirely relieved; the other remaining forty, who improve in different degrees. The analysis in the present work may, perhaps, explain why the percentage of failures is so high. It seems that no school of psychotherapists has analysed 'mental' ills from the general *non-el* structural and semantic point of view; and, although the physicians struggle in every case to abolish the relative unconditionality of the reactions, their methods are neither neurological, nor physiological, nor fundamental enough.

The language used in these scientific theories includes such terms as 'conscious', 'unconscious', 'repression', 'inhibition', 'transfer', 'complex',. There seems no doubt that some such terms cover a few of the facts we know from experience and observation, and that they may be structurally correct on the psycho-logical level. The nervous mechanism involved, although discovered twenty-five years ago, has not generally attracted the attention of physicians, and the postulated theories, lacking neurological foundations, are often called 'far-fetched speculations', a fact which is ultimately harmful to the whole psychotherapeutic and semantic hygiene movement.

The 'psychologists' and the psychiatrists are very much divided as to the role 'introspection' plays. This is due to the confusion of the orders of abstractions. Animals may 'feel', may 'suffer', but they cannot *describe*. Humans differ in this respect; the given person may feel pain, the pain is very *objective* to the given individual, and it is *not words (objective level);* but we can describe it, this description being valid on the *descriptive level,* a higher order abstraction than the objective level (which is un-speakable for the given individual). If we *ascribe* this process to others, this is no longer a description but an inference, or a still higher order abstraction, which statements have to be verified by averaging. Scientifically (1933), psycho-logics are impossible without the *description* of internal processes, and, therefore, some 'introspection', so that the United States Behaviourism becomes a very naïve discipline. The Behaviourists mean well, methodologically, without realizing fully what scientific methodology is. They completely condemn 'introspection', yet they continually use it. Consciousness of abstracting solves the riddles of pro- or anti-behaviouristic attitudes, because, when we are fully

conscious of abstracting, we should never confuse description with inference, neurologically processes of different order.

Any discipline, to be a 'science', must start with the lowest abstractions available; which means descriptions of some objective, *un-speakable* level. In *human* psycho-logics, 'introspection' is the only *possible descriptive level,* all other methods being inferential.

The experiments of Doctor Zavadzki were conducted to investigate the mechanism of the so-called 'delayed reflexes'. Speaking roughly, in experiments in which the interval between the conditional stimulus and the reinforcement by food or acid is short, say, one to five seconds, the salivary secretion follows nearly immediately after the application of the conditional stimulus. If the delay between the two is longer, say, several minutes, the appearance of salivary secretions is also delayed, the length of this delay being proportional to the length of the interval between the two stimuli.

In these experiments there were two phases: the one in which the conditional stimulus has apparently no effect; the other in which the conditional stimulus becomes effective. The investigation was continued to discover what becomes of the excitation due to the conditional stimulus during its apparent inactivity.

New experiments finally disclosed an astonishing mechanism. A tactile stimulation was used for three minutes as a conditional stimulus for acid, and reinforced, as usual, by the application of acid, and a stable, delayed conditional reaction was obtained. But when a perfectly neutral stimulus, say, the sound of a metronome or a noiselessly rotating object, never connected with any alimentary stimulation, was superimposed upon the original conditional stimulus, immediately a copious secretion of saliva, together with the motor reactions peculiar to a given stimulus, were obtained.

We see that the excitatory process in the nervous system existed all the time in a concealed, non-manifest form and was released by an extra and neutral stimulus.[1]

Similar experiments show clearly that the structure and function of the central nervous system is such that some stimulations can be concealed and become macroscopically seemingly inactive, giving no obvious manifestation or response, yet preserving their active exciting characteristics which, by proper treatment, can be released at will. In physics, we have a similar phenomenon in the case of 'frozen' light, galvanic and storage batteries, pear-shaped drops of glass resulting from melting, which explode when the end is broken off, and many others, although probably the sub-microscopic mechanisms are different.

It does not take much explanation to see that the nervous mechanism revealed in the experiments of Doctor Zavadzki accounts on human levels for a great many 'mental' manifestations, including 'recall', 'unconscious', 'repression', 'complexes'., and allowing a further generalization, that a slight nervous disturbance of 'recall', in the sense of negative unconditionality, may be closely connected with a pathological semantic 'complex'.

Another experiment has close connection with the problems of the human 'unconscious', 'repressions', and 'complexes'. The positive conditional reactions were usually obtained by combining under certain conditions a formerly neutral stimulus with food or with a mild defense reaction to acid. If the neutral stimulus is not reinforced, it loses its significance for the organism, no secretion is obtained, and it becomes from this point of view a negative stimulus. If, with a given animal, a negative reaction is established, it can, under certain conditions, be transformed into a positive one by reinforcement. In the experiment we are describing, a dog was used, with a well-established negative alimentary reaction to the beats of the metronome at the rate of sixty beats per minute, while the rate of one hundred and twenty beats per minute was used as a positive stimulus. Both reactions were constant and precise. The process of transformation from negative to positive went slowly; after the seventeenth application with reinforcement, a small secretion of saliva was obtained; after the twenty-seventh reinforcement, the secretions of saliva were already considerable. No definite disturbances in other positive reactions were observed except for a tendency to the equalization of strong and weak conditional stimuli.

But the secretory reaction to the transformed stimulus of sixty beats did not remain constant, in spite of reinforcement; it diminished, and after the thirtieth application fell to zero. It was noticed, further, that immediately after the application of the metronome at the rate of sixty beats per minute, practically all the older positive reactions disappeared. After further experimenting, some of the positive effects of the metronome at sixty returned, but its negative or depressing effect on the positive reactions persisted. In all cases where the metronome at sixty was not used, all the positive conditional reactions maintained their strength, except that the weaker stimuli had an inclination to produce lesser effects toward the end of the experiment. Although the metronome at sixty or one hundred and twenty produced salivary secretions in varying quantities when used alone, whenever the metronome was used there followed a disturbance of all conditional reactions, varying from complete extinction to a diminution of secretions. The formerly positive stimulus

of one hundred and twenty beats of the metronome produced greater disturbances than the formerly negative of sixty. Further experimentation disclosed that the disturbance of the cortex was profound, and that it could not withstand any kind of stronger stimuli without producing completely negative results. It became, also, obvious that the maximum disturbance in the central nervous activity in animals (and in man) does not appear immediately after the application of the injurious factor but after a delay.

Since other auditory stimuli acted during these experiments, Pavlov concludes that 'the disturbance must be regarded as a result of a strictly localized functional interference in the acoustic analyser, a chronic functional lesion of some circumscribed part, the stimulation of which produces an immediate effect upon the function of the whole cortex, and finally leads to a protracted pathological state', and, that 'it is obvious that the localized disturbance of the acoustic analyser is again the result of a clash between excitation and inhibition', which this particular nervous system finds difficult to adjust.[2]

These experiments were conducted upon a dog which had served in the laboratory for a long time and belonged to the type which has a very negatively excitable nervous system. Experiments on dogs with very positively excitable nervous systems, although different in details, led to similar general results; namely, that a clashing of the two antagonistic nervous processes led, usually, to a more or less protracted disturbance of the function of the cortex, in the form of a lasting predominance of one of the processes.[3]

Experimenting on the conditional reactions in animals, such as a dog, by inducing pathological states of the nervous system, gives us, in a *simplified form,* a means to understand the mechanism which underlies some of the human 'mental' illnesses, provided we realize the fundamental fact that these experiments on dogs correspond, in their simpler form, to 'mental' ills, and not to 'mental' health, in man. The above experiments would be impossible with a healthy person; yet they depict exactly what happens in the case of 'mentally' ill. The experiments started with a healthy animal, and they ended with a pathological case. If similar experiments were undertaken with a healthy person, no pathological results would follow, owing to the larger conditionality of reactions.; but similar pathological results are produced in humans by different means, the confusion of orders of abstractions being a standard semantic mechanism to bring about the 'clash' between the positive and negative excitations which the nervous system of man cannot resolve so easily.

The experiments of Pavlov disclose, also, a fact which, on human levels, introduces serious complications; namely, that some animals have highly excitable nervous systems, and that some have less excitable ones. Experiments conducted on some individuals produce one effect; similar experiments performed on individuals with different nervous systems produce different results. In some instances, the nervous systems are so sturdy that no disturbances appear at all.

To anticipate a little: it appears that under the present linguistic, educational, social, economic., conditions, nearly all of us suffer from nervous and semantic disturbances, produced by copying animals in our nervous responses. This last condition occurs because the larger conditionality of human responses has not been taken into consideration; its mechanism is unknown, and we do our best to teach and enforce the animalistic responses. As yet, we have had no physiological and simple methods by which to train in this larger conditionality. This is a simple explanation of our failure. Only a few of us have such sturdy nervous systems that they do not become semantically disturbed in any marked degree, and these are exceptions. Obviously, even an attempt to build a general theory dealing with these semantic problems may be useful, for the very mistakes made may serve others as an incentive for further enquiries in a field which is practically unexplored and extremely large.

In the formulation of the present general theory, theoretical considerations suggested necessary neurological mechanisms; yet the standard books on physiology and neurology did not give enough data. In the recent work of Pavlov, I found sufficient experiments and data to illustrate the neurological mechanisms which underlie the present theory. It seems likely that the work of Pavlov and the experiments described, together with the theoretical issues raised in the present system, will be of value to psycho-logicians and psychiatrists, provided that they pay attention to the semantic non-confusion of orders of abstractions, without which it is practically impossible to translate experiments dealing with nervous responses of animals to the human levels and escape verbal fallacies. The language of the structure, as introduced in the present work, is essential in this respect; in fact, the present writer could not have carried out his analysis without it.

Pavlov also suggests some applications to human pathological cases which are recognized as such, the average person being assumed 'normal'. The present work is an independent theoretical enquiry, and the results are much more general, as they show that the general neurological mechanism allows an almost universal misuse of our nervous systems, because of the disregard of structural, linguistic, and semantic issues.

In several chapters, Pavlov discusses a large number of experiments dealing with functionally induced pathological states of the nervous system, and suggests, also, some therapeutic measures. He concludes: 'This . . . and other observations suggest that a gradual development of internal inhibition in the cortex should be used for re-establishment of the balance of normal conditions in cases of an unbalanced nervous system . . . I do not know whether similar therapeutic measures . . . are applied in human neurotherapy.'[4]

The above remark is vitally important; it is not only a result of a lifetime of scientific work, but it expresses a principle which is used, *without being formulated explicitly,* all through psychotherapy. In the present volume, this principle is not only formulated in physiological terms, but is also made the foundation of a physiological method for its semantic application. This method is found in the training and development of the consciousness of abstracting (see Part VII), which, when applied, not only restores nervous balance as empirically shown, but also gives powerful *preventive semantic means* if used in early education.

Further consequences and conclusions are given in Part VII. At this point, we shall merely state that the above explanations also show why a theory of *universal agreement,* in the broadest sense; namely, agreement with one's self, eliminating internal 'conflict', and with others, eliminating family, social, and international conflicts., is neurologically not only possible, but also a necessary semantic consequence of using the human nervous system in its structurally appropriate way.

It is well known that the use of the terms 'positive' and 'negative' is optional; but the opposing character of the issues involved is not optional, because these are experimental and structural. In former days, we not only made our selection, and called some issues positive and some negative, but we naturally had and have some semantic responses connected with them. Thus something 'positive' implied certainty, 'reality', 'truth', 'absolute'.; something 'negative' implied the negation of these.

In 1933, it appears likely that we shall have to revise *in toto* these semantic orientations, which obviously we have been practicing since the days of savagery.

What are the facts? Curiously enough:

1) The electricity which lights our lamps or runs our dynamos, we call, in the old language, 'negative' electricity.

2) The numbers which are the foundation of the most important complex numbers in mathematics are formally based on negative numbers.

3) The foundation of so-called 'human mentality' is the 'negative' reaction.

4) Because words are never the things we speak about, the sole link between languages and the objective world being structural, the only 'positive' facts about this world are of the old 'negative' character.

5) Finally, the main difficulty of the A-system can be found in the positive 'is' of identity, involving us in false to fact evaluation and semantic disturbances; on the other hand, a $\bar{A}$-system is based on the complete elimination of identity formulated as a negative premise of the 'this *is not* this' type (see Part VII).

At present, only in *technical* mathematics can people behave semantically like 'gentlemen'. They analyse and agree; no quarrels are possible. Linguistic and semantic researches show that the structure of all languages can, and *must,* be made similar to empirical structures; and then, also, the rest of humans can, and probably will, behave in a less silly and futile way than they have done in the past and are doing in the present.